National Taiwan University

Tideology

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Contest (1)

todo.txt

4 lines

```
code run
chmod +x run
code tem.cpp
./run tem
```

template.cpp

Description: Contest template. Debug with ./run (defines tsao); check typing with ./hash template.

d7c087, 49 lines

```
#include <bits/stdc++.h>
#include <bits/extc++.h>
using namespace std;
#define int long long
#define all(x) (x).begin(), (x).end()
#define pb push_back
#define F first
#define S second
#define pii pair<int, int>
#define rep(i, a, b) for(int i = a; i < (b); ++i)
#define sz(x) (int)(x).size()
#define vi vector<int>
#define vll vector<ll>
#define pll pair<ll, ll>
typedef long long ll;
#ifdef tsao
template <typename A, typename B>
ostream& operator << (ostream &o, const pair<A, B> &p){
  return o << "(" << p.F << ", " << p.S << ")";
}
template <typename T>
ostream& operator << (ostream &o, const vector<T> &v){
  o << "{"; int f = 0;
  for(auto &i : v) o << (f++ ? " " : "") << i;
  return o << "}";
}
void bug__(int c, auto ...a){
  cerr << "\e[1;" << c << "m";
  (..., (cerr << a << " "));
  cerr << "\e[0m\n";
}
#define bug_(c, x...) bug__(c, __LINE__, "[" + string(#x) + "]", x)
#define bug(x...) bug_(32, x)
#define bugv(x...) bug_(36, vector(x))
#define safe bug_(33, "safe")
#else
#define bug(x...) void(0)
#define bugv(x...) void(0)
#define safe void(0)
#endif
void solve(){

}
signed main(){
  ios::sync_with_stdio(0);
  cin.tie(0); cout.tie(0);
  int t = 1;
  while(t--) solve();
}
```

fast.cpp

26 lines

```
#pragma GCC optimize("Ofast,no-stack-protector")
#pragma GCC optimize("no-math-errno,unroll-loops")
#pragma GCC target("sse,sse2,sse3,ssse3,sse4")
#pragma GCC target("popcnt,abm,mmx,avx,arch=skylake")
// in main: __builtin_ia32_ldmxcsr(__builtin_ia32_stmxcsr()|0x8040);
#include<unistd.h>
char OB[65536]; int OP;
struct Fl{ ~Fl(){ write(1, OB, OP); } } fl_;
inline int RC() {
  static char buf[65536], *p = buf, *q = buf;
  return p == q && (q = (p = buf) + read(0, buf, 65536)) == buf
      ? -1 : *p++;
}
inline int R() {
  int c = RC(), s = 1, a = 0;
  while(c != '-' && c < '0' && c != -1) c = RC();
  if(c == '-') s = -1, c = RC();
  for(; c >= '0'; c = RC()) a = a * 10 + (c ^ '0');
  return a * s;
}
inline void W(int n) {
  static char buf[20]; int p = 0;
  if (n < 0) OB[OP++] = '-', n = -n;
  do buf[p++] = '0' + n % 10, n /= 10; while (n);
  while (p) OB[OP++] = buf[--p];
  if (OP > 65500) write(1, OB, OP), OP = 0;
}
```

run

12 lines

```
F="-std=gnu++20 -Dtsao -Wall -Wextra -Wshadow -g -O2 -pipe"
F="$F -fsanitize=address,undefined"
F="$F -fno-sanitize-recover=all -D_GLIBCXX_ASSERTIONS"
if [ ! -f bits/stdc++.h.gch ]; then # precompile, once
  mkdir -p bits
  echo '#include_next <bits/stdc++.h>' > bits/stdc++.h
  echo '#include <bits/stdc++.h>' > pch.h
  echo '#include <bits/extc++.h>' >> pch.h
  g++ $F -x c++-header pch.h -o bits/stdc++.h.gch
fi
export ASAN_OPTIONS=detect_leaks=0:handle_abort=1
g++ $F -I. $1.cpp -o $1 && ./$1
```

s.sh

11 lines

```
#!/usr/bin/env bash
for f in gen good bad; do g++ -std=c++20 -O2 $f.cpp -o $f ||
    exit; done
i=1
while :; do
  echo $i
  ./gen $i > in
  ./good < in > out1
  timeout 2 ./bad < in > out2 || { echo "RE/TLE"; break; }
  diff out1 out2 || break
  ((++i))
done
```

hash

1 lines

```
cpp -w -dD -P -fpreprocessed $1.cpp | tr -d '[:space:]' |
    md5sum | cut -c-6
```

Mathematics (2)

2.1 Equations

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The extremum is given by $x = -b/2a$.

$$\begin{matrix} ax + by = e \\ cx + dy = f \end{matrix} \Rightarrow \begin{matrix} x = \frac{ed - bf}{ad - bc} \\ y = \frac{af - ec}{ad - bc} \end{matrix}$$

In general, given an equation $Ax = b$, the solution to a variable x_i is given by

$$x_i = \frac{\det A_i'}{\det A}$$

where A_i' is A with the i'th column replaced by b.

2.2 Recurrences

If $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$, and $r_1, \ldots, r_k$ are distinct roots of $x^k - c_1 x^{k-1} - \cdots - c_k$, there are $d_1, \ldots, d_k$ s.t.

$$a_n = d_1 r_1^n + \cdots + d_k r_k^n.$$

Non-distinct roots r become polynomial factors, e.g. $a_n = (d_1 n + d_2) r^n$.

2.3 Trigonometry

$$\sin(v + w) = \sin v \cos w + \cos v \sin w$$
$$\cos(v + w) = \cos v \cos w - \sin v \sin w$$
$$\tan(v + w) = \frac{\tan v + \tan w}{1 - \tan v \tan w}$$
$$\sin v + \sin w = 2 \sin \frac{v + w}{2} \cos \frac{v - w}{2}$$
$$\cos v + \cos w = 2 \cos \frac{v + w}{2} \cos \frac{v - w}{2}$$
$$(V + W) \tan(v - w)/2 = (V - W) \tan(v + w)/2$$

where V, W are lengths of sides opposite angles v, w.

$$a \cos x + b \sin x = r \cos(x - \phi)$$
$$a \sin x + b \cos x = r \sin(x + \phi)$$

where $r = \sqrt{a^2 + b^2}, \phi = \text{atan2}(b, a)$.

2.4 Geometry

2.4.1 Triangles

Side lengths: a, b, c

Semiperimeter: $p = \frac{a+b+c}{2}$

Area: $A = \sqrt{p(p-a)(p-b)(p-c)}$

Circumradius: $R = \frac{abc}{4A}$

Inradius: $r = \frac{A}{p}$

Length of median (divides triangle into two equal-area triangles): $m_a = \frac{1}{2}\sqrt{2b^2+2c^2-a^2}$

Length of bisector (divides angles in two):

$$s_a = \sqrt{bc\left[1-\left(\frac{a}{b+c}\right)^2\right]}$$

Law of sines: $\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c} = \frac{1}{2R}$

Law of cosines: $a^2 = b^2 + c^2 - 2bc\cos\alpha$

Law of tangents: $\frac{a+b}{a-b} = \frac{\tan\frac{\alpha+\beta}{2}}{\tan\frac{\alpha-\beta}{2}}$

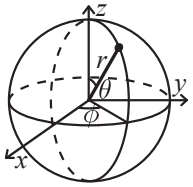
2.4.2 Quadrilaterals

With side lengths a, b, c, d, diagonals e, f, diagonals angle θ, area A and magic flux $F = b^2 + d^2 - a^2 - c^2$:

$$4A = 2ef \cdot \sin\theta = F\tan\theta = \sqrt{4e^2f^2 - F^2}$$

For cyclic quadrilaterals the sum of opposite angles is $180°$, $ef = ac + bd$, and $A = \sqrt{(p-a)(p-b)(p-c)(p-d)}$.

2.4.3 Spherical coordinates



$$x = r\sin\theta\cos\phi \qquad r = \sqrt{x^2+y^2+z^2}$$
$$y = r\sin\theta\sin\phi \qquad \theta = \operatorname{acos}(z/\sqrt{x^2+y^2+z^2})$$
$$z = r\cos\theta \qquad \phi = \operatorname{atan2}(y, x)$$

2.5 Derivatives/Integrals

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}\tan x = 1 + \tan^2 x \qquad \frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$
$$\int \tan ax = -\frac{\ln|\cos ax|}{a} \qquad \int x\sin ax = \frac{\sin ax - ax\cos ax}{a^2}$$
$$\int e^{-x^2} = \frac{\sqrt{\pi}}{2}\operatorname{erf}(x) \qquad \int xe^{ax}dx = \frac{e^{ax}}{a^2}(ax-1)$$

Integration by parts:

$$\int_a^b f(x)g(x)dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x)dx$$

2.6 Sums

$$c^a + c^{a+1} + \cdots + c^b = \frac{c^{b+1} - c^a}{c-1}, c \neq 1$$
$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$
$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(2n+1)(n+1)}{6}$$
$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$
$$1^4 + 2^4 + 3^4 + \cdots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

2.7 Series

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots, (-\infty < x < \infty)$$
$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots, (-1 < x \leq 1)$$
$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{2x^3}{32} - \frac{5x^4}{128} + \ldots, (-1 \leq x \leq 1)$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots, (-\infty < x < \infty)$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots, (-\infty < x < \infty)$$

2.8 Probability theory

Let X be a discrete random variable with probability $p_X(x)$ of assuming the value x. It will then have an expected value (mean) $\mu = \mathbb{E}(X) = \sum_x xp_X(x)$ and variance $\sigma^2 = V(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_x (x - \mathbb{E}(X))^2 p_X(x)$ where σ is the standard deviation. If X is instead continuous it will have a probability density function $f_X(x)$ and the sums above will instead be integrals with $p_X(x)$ replaced by $f_X(x)$.

Expectation is linear:

$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$$

For independent X and Y,

$$V(aX + bY) = a^2V(X) + b^2V(Y).$$

Data structures (3)

OrderStatisticTree.h

Description: A set (not multiset!) with support for finding the n'th For multiset, change less to less_equal, but lower_bound will become upper_bound and you can only erase iterator element, and finding the index of an element. To get a map, change `null_type`.
Time: $\mathcal{O}(\log N)$

819d08, 15 lines

```
using namespace __gnu_pbds;

template<class T>
using Tree = tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;
void example(){
  Tree<int> t, t2; t.insert(8);
  auto it = t.insert(10).first;
  assert(it == t.lower_bound(9));
  assert(t.order_of_key(10) == 1);
  assert(t.order_of_key(11) == 2);
  assert(*t.find_by_order(0) == 8);
  t.join(t2); // assuming T< T2 or T> T2, merge t2 into t
}
```

HashMap.h

Description: Hash map with mostly the same API as unordered_map, but ~3x faster. Uses 1.5x memory. Initial capacity must be a power of 2 (if provided).

140fd8, 6 lines

```
// To use most bits rather than just the lowest ones:
struct chash { // large odd number for C
  const uint64_t C = (ll)(4e18 * acos(0)) | 71;
  ll operator()(ll x) const { return __builtin_bswap64(x*C); }
};
__gnu_pbds::gp_hash_table<ll, int, chash> h({}, {}, {}, {}, {
    1<<16});
```

Matrix.h

Description: Basic operations on square matrices.
Usage:

```
Matrix<int, 3> A;
A.d = {{{1, 2, 3}}, {{4, 5, 6}}, {{7, 8, 9}}};
array<int, 3> vec = {1, 2, 3};
vec = (A^N) * vec;
```

faa329, 26 lines

```
template<class T, int N> struct Matrix {
  typedef Matrix M;
  array<array<T, N>, N> d{};
  M operator*(const M& m) const {
    M a;
    rep(i, 0, N) rep(k, 0, N)
      rep(j, 0, N) a.d[i][j] += d[i][k]*m.d[k][j];
    return a;
  }
  array<T, N> operator*(const array<T, N>& vec) const {
    array<T, N> ret{};
    rep(i, 0, N) rep(j, 0, N) ret[i] += d[i][j] * vec[j];
    return ret;
  }
  M operator^(ll p) const {
    assert(p >= 0);
    M a, b(*this);
    rep(i, 0, N) a.d[i][i] = 1;
    while(p){
      if(p&1) a = a*b;
      b = b*b;
      p >>= 1;
    }
    return a;
  }
};
```

LineContainer.h

Description: Container where you can add lines of the form kx+m, and query maximum values at points x. Useful for dynamic programming ("convex hull trick").
Time: $\mathcal{O}(\log N)$

8ec1c7, 30 lines

```
struct Line {
  mutable ll k, m, p;
  bool operator<(const Line& o) const { return k < o.k; }
  bool operator<(ll x) const { return p < x; }
};
```

```
struct LineContainer : multiset<Line, less<>> {
  // (for doubles, use inf = 1/.0, div(a, b) = a/b)
  static const ll inf = LLONG_MAX;
  ll div(ll a, ll b){ // floored division
    return a / b - ((a ^ b) < 0 && a % b); }
  bool isect(iterator x, iterator y){
    if(y == end()) return x->p = inf, 0;
    if(x->k == y->k) x->p = x->m > y->m ? inf : -inf;
    else x->p = div(y->m - x->m, x->k - y->k);
    return x->p >= y->p;
  }
  void add(ll k, ll m){
    auto z = insert({k, m, 0}), y = z++, x = y;
    while(isect(y, z)) z = erase(z);
    if(x != begin() && isect(--x, y)) isect(x, y = erase(y));
    while((y = x) != begin() && (--x)->p >= y->p)
      isect(x, erase(y));
  }
  ll query(ll x){
    assert(!empty());
    auto l = *lower_bound(x);
    return l.k * x + l.m;
  }
};
```

Lichao.h

Description: Li Chao tree on known x-coordinates; supports line/segment add.
Usage: `LiChao tr(xs); tr.add({m, b}); tr.add({m, b}, l, r); tr.query(x);`
Time: $\mathcal{O}(logN)$ per line/query, $\mathcal{O}\left(log^2 N\right)$ per segment

23b102, 42 lines

```
struct Line {
  ll m=0, b=0;
  ll operator()(ll x) const { return m*x+b; }
};
struct LiChao {
  const ll INF = 4e18;
  vector<ll> x; vector<Line> st;
  bool mn;
  LiChao(vector<ll> x, bool mn=true) : x(x), mn(mn){
    sort(all(this->x)); this->x.erase(unique(all(this->x)),
        this->x.end());
    assert(sz(this->x));
    st.assign(4*sz(this->x), {0, mn?INF:-INF});
  }
  bool better(Line a, Line b, int p){ return mn ? a(x[p])<b(x[p
      ]) : a(x[p])>b(x[p]); }
  void add(Line f, int l, int r, int p){
    int m=(l+r)/2; bool a=better(f, st[p], l), b=better(f, st[p
        ], m);
    if(b) swap(f, st[p]);
    if(r-l==1) return;
    if(a!=b) add(f, l, m, p*2); else add(f, m, r, p*2+1);
  }
  void add(Line f){ add(f, 0, sz(x), 1); }
  void add(Line f, int a, int b, int l, int r, int p){
    if(b<=l or r<=a) return;
    if(a<=l and r<=b) return add(f, l, r, p);
    int m=(l+r)/2;
    add(f, a, b, l, m, p*2); add(f, a, b, m, r, p*2+1);
  }
  void add(Line f, ll l, ll r){ // x in [l, r]
    int a=lower_bound(all(x), l)-x.begin(), b=upper_bound(all(x
        ), r)-x.begin();
    if(a<b) add(f, a, b, 0, sz(x), 1);
  }
  ll query(int p, int l, int r, int q){
    ll ans=st[p](x[q]);
    if(r-l==1) return ans;
    int m=(l+r)/2; ll z=q<m?query(p*2, l, m, q):query(p*2+1, m,
        r, q);
    return mn?min(ans, z):max(ans, z);
  }
  ll query(ll q){
    q=lower_bound(all(x), q)-x.begin(); assert(q<sz(x));
    return query(1, 0, sz(x), q);
  }
};
```

Treap.h

Description: A short self-balancing tree. It acts as a sequential container with log-time splits/joins, and is easy to augment with additional data.
Time: $\mathcal{O}(\log N)$

9cf53d, 53 lines

```
struct Node {
  Node *l = 0, *r = 0;
  int val, y, c = 1;
  Node(int val) : val(val){ static mt19937 r; y = r(); }
  void recalc();
};

int cnt(Node* n){ return n ? n->c : 0; }
void Node::recalc(){ c = cnt(l) + cnt(r) + 1; }

template<class F> void each(Node* n, F f){
  if(n){ each(n->l, f); f(n->val); each(n->r, f); }
}

pair<Node*, Node*> split(Node* n, int k){
  if(!n) return {};
  if(cnt(n->l) >= k){ // "n->val >= k" for lower_bound(k)
    auto [L, R] = split(n->l, k);
    n->l = R;
    n->recalc();
    return {L, n};
  }else{
    auto [L, R] = split(n->r, k - cnt(n->l) - 1); // and just "
        k"
    n->r = L;
    n->recalc();
    return {n, R};
  }
}

Node* merge(Node* l, Node* r){
  if(!l) return r;
  if(!r) return l;
  if(l->y > r->y){
    l->r = merge(l->r, r);
    return l->recalc(), l;
  }else{
    r->l = merge(l, r->l);
    return r->recalc(), r;
  }
}

Node* ins(Node* t, Node* n, int pos){
  auto [l, r] = split(t, pos);
  return merge(merge(l, n), r);
}

// Example application: move the range [l, r) to index k
void move(Node*& t, int l, int r, int k){
  Node *a, *b, *c;
  tie(a, b) = split(t, l); tie(b, c) = split(b, r - l);
  if(k <= l) t = merge(ins(a, b, k), c);
  else t = merge(a, ins(c, b, k - r));
}
```

pre-seg.h

Description: Persistent segment tree: point assign and range sum on [0, n).
Usage: `Seg st(n); int root2=st.modify(root1, p, x); st.query(root2, l, r);`
Time: $\mathcal{O}(logn)$ per operation

b8993e, 27 lines

```
struct Seg {
  struct Node { int ch[2]{}; ll val=0; };
  vector<Node> t{{}};
  int n;
  Seg(int n, int q = 0) : n(n){ if(q) t.reserve(1+q*(__lg(n)+1)
      ); }
  int modify(int u, int l, int r, int p, ll x){
    int v = sz(t); t.pb(t[u]);
    if(r-l == 1) return t[v].val=x, v;
    int m=(l+r)/2, d=p>=m;
    if(d) t[v].ch[1]=modify(t[v].ch[1], m, r, p, x);
    else t[v].ch[0]=modify(t[v].ch[0], l, m, p, x);
    t[v].val=t[t[v].ch[0]].val+t[t[v].ch[1]].val;
    return v;
  }
  int modify(int root, int p, ll x){
    assert(0<=p and p<n); return modify(root, 0, n, p, x);
  }
  ll query(int u, int l, int r, int a, int b) const {
    if(!u or b<=l or r<=a) return 0;
    if(a<=l and r<=b) return t[u].val;
    int m=(l+r)/2;
    return query(t[u].ch[0], l, m, a, b)+query(t[u].ch[1], m, r
        , a, b);
  }
  ll query(int root, int l, int r) const {
    return query(root, 0, n, l, r);
  }
};
```

MoQueries.h

Description: Mo ordering for half-open intervals. add/del get (index, end).
Time: $\mathcal{O}(Nsqrt(Q))$ moves

bef3d1, 54 lines

```
template<class Add, class Del, class Calc>
auto mo(vector<pii> Q, Add add, Del del, Calc calc, int blk=0){
  if(!blk){ int R = 1; for(auto [l, r] : Q) R = max(R, r); blk
      = max<int>(1, R / sqrt(sz(Q) + 1)); }
  using T=decltype(calc());
  int L=0, R=0; vi ord(sz(Q)); vector<T> ans(sz(Q));
  iota(all(ord), 0);
  sort(all(ord), [&](int i, int j){
    auto [a, b]=Q[i]; auto [c, d]=Q[j];
    return pii(a/blk, b^-(a/blk&1)) < pii(c/blk, d^-(c/blk&1));
  });
  for(int i:ord){
    auto [l, r]=Q[i];
    while(L>l) add(--L, 0);
    while(R<r) add(R++, 1);
    while(L<l) del(L++, 0);
    while(R>r) del(--R, 1);
    ans[i]=calc();
  }
  return ans;
}

template<class Add, class Del, class Calc>
auto moTree(vector<array<int, 2>> Q, vector<vi> &g, Add add,
    Del del,
        Calc calc, int root=0, int blk=350){
  using T=decltype(calc());
```

```
  int n=sz(g), ti=n, pos[2]={root, root};
  vi ord(sz(Q)), I(n*2), L(n), R(n), in(n), par(n), tmp(n);
  vector<T> ans(sz(Q)); add(root, 0), in[root]=1;
  auto dfs=[&](auto self, int u, int p, int d)->void {
    par[u]=p; L[u]=ti;
    if(d) I[u]=ti++;
    for(int v:g[u]) if(v!=p) self(self, v, u, !d);
    if(!d) I[u]=ti++;
    R[u]=ti;
  };
  dfs(dfs, root, -1, 0); iota(all(ord), 0);
  sort(all(ord), [&](int i, int j){
    auto a=Q[i], b=Q[j];
    return pii(I[a[0]]/blk, I[a[1]]^-(I[a[0]]/blk&1)) <
           pii(I[b[0]]/blk, I[b[1]]^-(I[b[0]]/blk&1));
  });
  for(int qi:ord) rep(e, 0, 2){
    int &a=pos[e], b=Q[qi][e], k=0;
    auto step=[&](int c){
      if(in[c]) del(a, e), in[a]=0; else add(c, e), in[c]=1;
      a=c;
    };
    while(not(L[b]<=L[a] and R[a]<=R[b])) tmp[k++]=b, b=par[b];
    while(a!=b) step(par[a]);
    while(k) step(tmp[--k]);
    if(e) ans[qi]=calc();
  }
  return ans;
}
```

SqrtSum.h

Description: Sqrt decomposition: point add and range sum on [0, n). Pair with Mo (many adds, few queries), e.g. on the value domain. kth: min p with sum [0, p] > k, needs all a >= 0; n if none.
Usage: `SqrtSum s(n); s.add(i, v); s.query(l, r) = sum [l, r); s.kth(k);`
Time: $\mathcal{O}(1)$ add, $\mathcal{O}\left(\sqrt{N}\right)$ query/kth

e6ccad, 18 lines

```
struct SqrtSum {
  int n, B; vector<ll> a, blk;
  SqrtSum(int n) : n(n), B((int)sqrt(n)+1), a(n), blk(n/B+1){}
  void add(int i, ll v){ a[i]+=v, blk[i/B]+=v; } // a[i] += v
  ll query(int l, int r){
    ll s=0;
    for(; l<r and l%B; l++) s+=a[l];
    for(; l+B<=r; l+=B) s+=blk[l/B];
    for(; l<r; l++) s+=a[l];
    return s;
  }
  int kth(ll k){
    int b=0, i;
    for(; b<sz(blk) and blk[b]<=k; b++) k-=blk[b];
    for(i=b*B; i<n and a[i]<=k; i++) k-=a[i];
    return min(i, n);
  }
};
```

Numerical (4)

4.1 Polynomials and recurrences

Polynomial.h

c9b7b0, 17 lines

```
struct Poly {
  vector<double> a;
  double operator()(double x) const {
    double val = 0;
    for(int i = sz(a); i--;) (val *= x) += a[i];
    return val;
  }
  void diff(){
    rep(i, 1, sz(a)) a[i-1] = i*a[i];
    a.pop_back();
  }
  void divroot(double x0){
    double b = a.back(), c; a.back() = 0;
    for(int i=sz(a)-1; i--;) c = a[i], a[i] = a[i+1]*x0+b, b=c;
    a.pop_back();
  }
};
```

PolyRoots.h

Description: Finds the distinct real roots of a polynomial in [xmin, xmax].
Time: $\mathcal{O}\left(n^2 log(1/epsilon)\right)$

"Polynomial.h" 7f59a3, 28 lines

```
vector<double> polyRoots(Poly p, double xmin, double xmax){
  while(sz(p.a) > 1 && abs(p.a.back()) < 1e-14) p.a.pop_back();
  if(sz(p.a) <= 1) return {};
  if(sz(p.a) == 2){
    double x = -p.a[0]/p.a[1];
    return xmin <= x && x <= xmax ? vector<double>{x} : vector<
        double>{};
  }
  Poly der = p; der.diff();
  auto at = polyRoots(der, xmin, xmax);
  at.insert(at.begin(), xmin); at.pb(xmax);
  vector<double> ret;
  auto add = [&](double x){
    if(abs(p(x)) < 1e-8 && (ret.empty() || abs(ret.back()-x) >
        1e-7)) ret.pb(x);
  };
  rep(i, 0, sz(at)){
    add(at[i]);
    if(i+1 == sz(at)) break;
    double l = at[i], r = at[i+1], fl = p(l), fr = p(r);
    if((fl > 0) == (fr > 0) || abs(fl) < 1e-8 || abs(fr) < 1e
        -8) continue;
    rep(it, 0, 80){
      double m = (l+r)/2, fm = p(m);
      if((fl > 0) == (fm > 0)) l = m, fl = fm;
      else r = m;
    }
    add((l+r)/2);
  }
  return ret;
}
```

PolyInterpolate.h

Description: Given n points (x[i], y[i]), computes an n-1-degree polynomial p that passes through them: $p(x) = a[0] * x^0 + ... + a[n-1] * x^{n-1}$. For numerical precision, pick $x[k] = c * \cos(k/(n-1) * \pi), k = 0 \ldots n-1$.
Time: $\mathcal{O}\left(n^2\right)$

08bf48, 13 lines

```
typedef vector<double> vd;
vd interpolate(vd x, vd y, int n){
  vd res(n), temp(n);
  rep(k, 0, n-1) rep(i, k+1, n)
    y[i] = (y[i] - y[k]) / (x[i] - x[k]);
  double last = 0; temp[0] = 1;
  rep(k, 0, n) rep(i, 0, n){
    res[i] += y[k] * temp[i];
    swap(last, temp[i]);
    temp[i] -= last * x[k];
  }
  return res;
}
```

BerlekampMassey.h

Description: Recovers any n-order linear recurrence relation from the first $2n$ terms of the recurrence. Useful for guessing linear recurrences after brute-forcing the first terms. Should work on any field, but numerical stability for floats is not guaranteed. Output will have size $\leq n$.
Usage: `berlekampMassey({0, 1, 1, 3, 5, 11}) // {1, 2}`
Time: $\mathcal{O}\left(N^2\right)$

"../number-theory/ModPow.h" 84669d, 20 lines

```
vector<ll> berlekampMassey(vector<ll> s){
  int n = sz(s), L = 0, m = 0;
  vector<ll> C(n + 1), B(n + 1), T;
  C[0] = B[0] = 1;

  ll b = 1;
  rep(i, 0, n){ ++m;
    ll d = s[i] % mod;
    rep(j, 1, L+1) d = (d + C[j] * s[i - j]) % mod;
    if(!d) continue;
    T = C; ll coef = d * modpow(b, mod-2) % mod;
    rep(j, m, n) C[j] = (C[j] - coef * B[j - m]) % mod;
    if(2 * L > i) continue;
    L = i + 1 - L; B = T; b = d; m = 0;
  }

  C.resize(L + 1); C.erase(C.begin());
  for(ll& x : C) x = (mod - x) % mod;
  return C;
}
```

LinearRecurrence.h

Description: Generates the k'th term of an n-order linear recurrence $S[i] = \sum_j S[i-j-1]tr[j]$, given $S[0 \ldots \geq n-1]$ and $tr[0 \ldots n-1]$. Faster than matrix multiplication. Useful together with Berlekamp–Massey.
Usage: `linearRec({0, 1}, {1, 1}, k) // k'th Fibonacci number`
Time: $\mathcal{O}\left(n^2 \log k\right)$

f4e444, 26 lines

```
typedef vector<ll> Poly;
ll linearRec(Poly S, Poly tr, ll k){
  int n = sz(tr);

  auto combine = [&](Poly a, Poly b){
    Poly res(n * 2 + 1);
    rep(i, 0, n+1) rep(j, 0, n+1)
      res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
    for(int i = 2 * n; i > n; --i) rep(j, 0, n)
      res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
    res.resize(n + 1);
    return res;
  };

  Poly pol(n + 1), e(pol);
  pol[0] = e[1] = 1;

  for(++k; k; k /= 2){
    if(k % 2) pol = combine(pol, e);
    e = combine(e, e);
  }

  ll res = 0;
  rep(i, 0, n) res = (res + pol[i + 1] * S[i]) % mod;
  return res;
}
```

Operation.h

Description: Polynomial operations from std_abs codebook. Uses NumberTheoreticTransform.h. Sqrt needs QuadraticResidue(a, mod) returning -1 if none (ModSqrt.h asserts instead); TaylorShift/SamplingShift need fac[i] = i!, facp[i] = 1/i!.

"NumberTheoreticTransform.h" 3f5ca2, 185 lines

```
typedef vector<ll> Poly;
Poly Mul(Poly a, Poly b, int bound = N) {
  int m = sz(a) + sz(b) - 1, n = 1;
  while (n < m) n <<= 1;
  a.resize(n), b.resize(n);
  ntt(a), ntt(b);
  for (int i = 0; i < n; ++i) a[i] = mul(a[i], b[i]);
  ntt(a, true), a.resize(min(m, bound));
  return a;
} // 8e2e8b
Poly Inverse(Poly a) {
  // O(NlogN), a[0] != 0
  int n = sz(a);
  Poly res(1, Pow(a[0], mod - 2));
  for (int m = 1; m < n; m <<= 1) {
    if (n < m * 2) a.resize(m * 2);
    Poly v1(a.begin(), a.begin() + m * 2), v2 = res;
    v1.resize(m * 4), v2.resize(m * 4);
    ntt(v1), ntt(v2);
    for (int i = 0; i < m * 4; ++i)
      v1[i] = mul(mul(v1[i], v2[i]), v2[i]);
    ntt(v1, true);
    res.resize(m * 2);
    for (int i = 0; i < m; ++i)
      res[i] = add(res[i], res[i]);
    for (int i = 0; i < m * 2; ++i)
      res[i] = sub(res[i], v1[i]);
  }
  res.resize(n);
  return res;
} // 4d79c8
pair <Poly, Poly> Divide(Poly a, Poly b) {
  // a = bQ + R, O(NlogN), b.back() != 0
  int n = sz(a), m = sz(b), k = n - m + 1;
  if (n < m) return {{0}, a};
  Poly ra = a, rb = b;
  reverse(all(ra)), ra.resize(k);
  reverse(all(rb)), rb.resize(k);
  Poly Q = Mul(ra, Inverse(rb), k);
  reverse(all(Q));
  Poly res = Mul(b, Q), R(m - 1);
  for (int i = 0; i < m - 1; ++i)
    R[i] = sub(a[i], res[i]);
  return {Q, R};
} // 7d15e3
Poly SqrtImpl(Poly a) {
  if (a.empty()) return {0};
  int z = QuadraticResidue(a[0], mod), n = sz(a);
  if (z == -1) return {-1};
  Poly q(1, z);
  const int inv2 = (mod + 1) / 2;
  for (int m = 1; m < n; m <<= 1) {
    if (n < m * 2) a.resize(m * 2);
    q.resize(m * 2);
    Poly f2 = Mul(q, q, m * 2);
    for (int i = 0; i < m * 2; ++i)
      f2[i] = sub(f2[i], a[i]);
    f2 = Mul(f2, Inverse(q), m * 2);
    for (int i = 0; i < m * 2; ++i)
      q[i] = sub(q[i], mul(f2[i], inv2));
  }
  q.resize(n);
  return q;
} // 984549
Poly Sqrt(Poly a) {
  // O(NlogN), return {-1} if not exists
  int n = sz(a), m = 0;
  while (m < n && a[m] == 0) m++;
  if (m == n) return Poly(n);
  if (m & 1) return {-1};
  Poly s = SqrtImpl(Poly(a.begin() + m, a.end()));
  if (s[0] == -1) return {-1};
  Poly res(n);
  for (int i = 0; i < sz(s); ++i)
    res[i + m / 2] = s[i];
  return res;
} // d1acd7
Poly Derivative(Poly a) {
  int n = sz(a);
  Poly res(n - 1);
  for (int i = 0; i < n - 1; ++i)
    res[i] = mul(a[i + 1], i + 1);
  return res;
} // 001be0
Poly Integral(Poly a) {
  int n = sz(a);
  Poly res(n + 1);
  for (int i = 0; i < n; ++i)
    res[i + 1] = mul(a[i], Pow(i + 1, mod - 2));
  return res;
} // 6fc53d
Poly Ln(Poly a) {
  // O(NlogN), a[0] = 1
  int n = sz(a);
  if (n == 1) return {0};
  Poly d = Derivative(a);
  a.pop_back();
  return Integral(Mul(d, Inverse(a), n - 1));
} // 377d20
Poly Exp(Poly a) {
  // O(NlogN), a[0] = 0
  int n = sz(a);
  Poly q(1, 1);
  a[0] = add(a[0], 1);
  for (int m = 1; m < n; m <<= 1) {
    if (n < m * 2) a.resize(m * 2);
    Poly g(a.begin(), a.begin() + m * 2), h(all(q));
    h.resize(m * 2), h = Ln(h);
    for (int i = 0; i < m * 2; ++i)
      g[i] = sub(g[i], h[i]);
    q = Mul(g, q, m * 2);
  }
  q.resize(n);
  return q;
} // 525e8f
Poly PolyPow(Poly a, ll k) {
  int n = sz(a), m = 0;
  Poly ans(n, 0);
  while (m < n && a[m] == 0) m++;
  if (k && m && (k >= n || k * m >= n)) return ans;
  if (m == n) return ans[0] = 1, ans;
  int lead = m * k;
  Poly b(a.begin() + m, a.end());
  ll base = Pow(b[0], k), inv = Pow(b[0], mod - 2);
  for (int i = 0; i < n - m; ++i)
    b[i] = mul(b[i], inv);
  b = Ln(b);
  for (int i = 0; i < n - m; ++i)
    b[i] = mul(b[i], k % mod);
  b = Exp(b);
  for (int i = lead; i < n; ++i)
    ans[i] = mul(b[i - lead], base);
  return ans;
} // 429212
vector <ll> Evaluate(Poly a, vector <ll> x) {
  if (x.empty()) return {};
  int n = sz(x);
  vector <Poly> up(n * 2);
  for (int i = 0; i < n; ++i)
    up[i + n] = {sub(0, x[i]), 1};
  for (int i = n - 1; i > 0; --i)
    up[i] = Mul(up[i * 2], up[i * 2 + 1]);
  vector <Poly> down(n * 2);
  down[1] = Divide(a, up[1]).second;
  for (int i = 2; i < n * 2; ++i)
    down[i] = Divide(down[i >> 1], up[i]).second;
  Poly y(n);
  for (int i = 0; i < n; ++i) y[i] = down[i + n][0];
  return y;
} // 84b37b
Poly Interpolate(vector <ll> x, vector <ll> y) {
  int n = sz(x);
  vector <Poly> up(n * 2);
  for (int i = 0; i < n; ++i)
    up[i + n] = {sub(0, x[i]), 1};
  for (int i = n - 1; i > 0; --i)
    up[i] = Mul(up[i * 2], up[i * 2 + 1]);
  Poly a = Evaluate(Derivative(up[1]), x);
  for (int i = 0; i < n; ++i)
    a[i] = mul(y[i], Pow(a[i], mod - 2));
  vector <Poly> down(n * 2);
  for (int i = 0; i < n; ++i) down[i + n] = {a[i]};
  for (int i = n - 1; i > 0; --i) {
    Poly lhs = Mul(down[i * 2], up[i * 2 + 1]);
    Poly rhs = Mul(down[i * 2 + 1], up[i * 2]);
    down[i].resize(sz(lhs));
    for (int j = 0; j < sz(lhs); ++j)
      down[i][j] = add(lhs[j], rhs[j]);
  }
  return down[1];
} // 12c6e1
Poly TaylorShift(Poly a, ll c) {
  // return sum a_i(x + c)^i;
  // fac[i] = i!, facp[i] = inv(i!)
  int n = sz(a);
  for (int i = 0; i < n; ++i) a[i] = mul(a[i], fac[i]);
  reverse(all(a));
  Poly b(n);
  int w = 1;
  for (int i = 0; i < n; ++i)
    b[i] = mul(facp[i], w), w = mul(w, c);
  a = Mul(a, b, n), reverse(all(a));
  for (int i = 0; i < n; ++i) a[i] = mul(a[i],facp[i]);
  return a;
} // 096757
```

4.2 Optimization

GoldenSectionSearch.h

Description: Finds the argument minimizing the function f in the interval $[a, b]$ assuming f is unimodal on the interval, i.e. has only one local minimum and no local maximum. The maximum error in the result is eps. Works equally well for maximization with a small change in the code. See TernarySearch.h in the Various chapter for a discrete version.

Usage: `double func(double x){ return 4+x+.3*x*x; }`
`double xmin = gss(-1000, 1000, func);`

Time: $\mathcal{O}\left(\log((b-a)/\epsilon)\right)$

31d45b, 14 lines

```
double gss(double a, double b, double (*f)(double)){
  double r = (sqrt(5)-1)/2, eps = 1e-7;
  double x1 = b - r*(b-a), x2 = a + r*(b-a);
  double f1 = f(x1), f2 = f(x2);
  while(b-a > eps)
```

```
    if(f1 < f2){ //change to > to find maximum
      b = x2; x2 = x1; f2 = f1;
      x1 = b - r*(b-a); f1 = f(x1);
    }else{
      a = x1; x1 = x2; f1 = f2;
      x2 = a + r*(b-a); f2 = f(x2);
    }
  return a;
}
```

HillClimbing.h

Description: Poor man's optimization for unimodal functions 8eeeaf, 14 lines

```
typedef array<double, 2> P;

template<class F> pair<double, P> hillClimb(P start, F f){
  pair<double, P> cur(f(start), start);
  for(double jmp = 1e9; jmp > 1e-20; jmp /= 2){
    rep(j, 0, 100) rep(dx, -1, 2) rep(dy, -1, 2){
      P p = cur.second;
      p[0] += dx*jmp;
      p[1] += dy*jmp;
      cur = min(cur, make_pair(f(p), p));
    }
  }
  return cur;
}
```

Simplex.h

Description: Maximizes $c^T x$ subject to $Ax \leq b$ and $x \geq 0$.
Duality: min $b^T y$ subject to $A^T y \geq c$, $y \geq 0$; finite optima agree. Returns {value, x}; value is -inf if infeasible and inf if unbounded.
Time: $\mathcal{O}(NM * \#pivots)$, exponential in the worst case 1589b0, 65 lines

```
typedef double T;
typedef vector<T>vd;
typedef vector<vd>vvd;
const T eps = 1e-8, inf = 1 / .0;
#define MP make_pair
#define ltj(X) if(s == -1 or MP(X[j], N[j]) < MP(X[s], N[s])) s
    = j

struct LPSolver{
  int m, n;
  vi N, B;
  vvd D;
  LPSolver(const vvd &A, const vd &b, const vd &c):
    m(sz(b)), n(sz(c)), N(n + 1), B(m), D(m + 2, vd(n + 2)){
    rep(i, 0, m) rep(j, 0, n) D[i][j] = A[i][j];
    rep(i, 0, m) B[i] = n + i, D[i][n] = -1, D[i][n + 1] = b[i
        ];
    rep(j, 0, n) N[j] = j, D[m][j] = -c[j];
    N[n] = -1;
    D[m + 1][n] = 1;
  }
  void pivot(int r, int s){
    T *a = D[r].data(), inv = 1 / a[s];
    rep(i, 0, m + 2) if(i != r and abs(D[i][s]) > eps){
      T *b = D[i].data(), inv2 = b[s] * inv;
      rep(j, 0, n + 2) b[j] -= a[j] * inv2;
      b[s] = a[s] * inv2;
    }
    rep(j, 0, n + 2) if(j != s) D[r][j] *= inv;
    rep(i, 0, m + 2) if(i != r) D[i][s] *= -inv;
    D[r][s] = inv;
    swap(B[r], N[s]);
  }
  bool simplex(int phase){
    int x = m + phase - 1;
    for(;;){
      int s = -1;
      rep(j, 0, n + 1) if(N[j] != -phase) ltj(D[x]);
      if(D[x][s] >= -eps) return 1;
      int r = -1;
      rep(i, 0, m){
        if(D[i][s] <= eps) continue;
        if(r == -1 or MP(D[i][n + 1] / D[i][s], B[i]) <
          MP(D[r][n + 1] / D[r][s], B[r])) r = i;
      }
      if(r == -1) return 0;
      pivot(r, s);
    }
  }
  pair<T, vd> solve(){
    int r = 0;
    rep(i, 1, m) if(D[i][n + 1] < D[r][n + 1]) r = i;
    if(D[r][n + 1] < -eps){
      pivot(r, n);
      if(!simplex(2) or D[m + 1][n + 1] < -eps) return {-inf, {
          }};
      rep(i, 0, m) if(B[i] == -1){
        int s = 0;
        rep(j, 1, n + 1) ltj(D[i]);
        pivot(i, s);
      }
    }
    bool ok = simplex(1);
    vd x(n);
    rep(i, 0, m) if(B[i] < n) x[B[i]] = D[i][n + 1];
    return {ok ? D[m][n + 1] : inf, x};
  }
};
```

4.3 Matrices

Determinant.h

Description: Calculates determinant of a matrix. Destroys the matrix.
Time: $\mathcal{O}(N^3)$ 24b41d, 15 lines

```
double det(vector<vector<double>> a){
  int n = sz(a); double res = 1;
  rep(i, 0, n){
    int b = i;
    rep(j, i+1, n) if(fabs(a[j][i]) > fabs(a[b][i])) b = j;
    if(i != b) swap(a[i], a[b]), res *= -1;
    res *= a[i][i];
    if(res == 0) return 0;
    rep(j, i+1, n){
      double v = a[j][i] / a[i][i];
      if(v != 0) rep(k, i+1, n) a[j][k] -= v * a[i][k];
    }
  }
  return res;
}
```

IntDeterminant.h

Description: Calculates determinant using modular arithmetics. Modulos can also be removed to get a pure-integer version.
Time: $\mathcal{O}(N^3)$ 2dc1e6, 19 lines

```
ll det(vector<vector<ll>> a, ll mod){
  int n = sz(a); ll ans = 1;
  for(auto &r : a) for(ll &x : r) x = (x%mod+mod)%mod;
  rep(i, 0, n){
    rep(j, i+1, n){
      while(a[j][i] != 0){ // gcd step
        ll t = a[i][i] / a[j][i];
        if(t){
          rep(k, i, n) a[i][k] = (a[i][k]-a[j][k]*t)%mod;
        }
        swap(a[i], a[j]);
        ans *= -1;
      }
    }
    ans = ans * a[i][i] % mod;
    if(!ans) return 0;
  }
  return (ans + mod) % mod;
}
```

SolveLinear.h

Description: Solves A*x=b. Returns rank, x, or -1, if inconsistent.
Time: $\mathcal{O}(N^2 M)$ 9a2cdf, 37 lines

```
typedef vector<double>vd;
const double eps = 1e-12;

pair<int, vd> solveLinear(vector<vd>A, vd b, int m = -1){
  int n = sz(A), rank = 0, br = 0, bc = 0;
  if(n) m = sz(A[0]);
  else if(m == -1) m = 0;
  vi col(m);
  iota(all(col), 0);
  rep(i, 0, n){
    double v, bv = 0;
    rep(r, i, n) rep(c, i, m) if((v = abs(A[r][c])) > bv)
      br = r, bc = c, bv = v;
    if(bv <= eps){
      rep(j, i, n) if(abs(b[j]) > eps) return {-1, {}};
      break;
    }
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    rep(j, 0, n) swap(A[j][i], A[j][bc]);
    bv = 1 / A[i][i];
    rep(j, i + 1, n){
      double z = A[j][i] * bv;
      b[j] -= z * b[i];
      rep(k, i + 1, m) A[j][k] -= z * A[i][k];
    }
    rank++;
  }
  vd x(m);
  for(int i = rank; i--;){
    b[i] /= A[i][i];
    x[col[i]] = b[i];
    rep(j, 0, i) b[j] -= A[j][i] * b[i];
  }
  return {rank, x};
}
```

SolveLinear2.h

Description: Solves A*x=b. Undetermined x[i] are NaN. Returns rank, x, or -1, if inconsistent.
Time: $\mathcal{O}(NMmin(N, M))$
"SolveLinear.h" 46b7a2, 34 lines

```
pair<int, vd> solveLinear2(vector<vd>A, vd b, int m = -1){
  int n = sz(A), r = 0;
  if(n) m = sz(A[0]);
  else if(m == -1) m = 0;
  vi col;
  rep(c, 0, m){
    if(r == n) break;
    int p = r;
    rep(i, r, n) if(abs(A[i][c]) > abs(A[p][c])) p = i;
    if(abs(A[p][c]) <= eps) continue;
    swap(A[p], A[r]);
```

```
    swap(b[p], b[r]);
    double z = A[r][c];
    rep(j, c, m) A[r][j] /= z;
    b[r] /= z;
    rep(i, 0, n) if(i != r){
      z = A[i][c];
      rep(j, c, m) A[i][j] -= z * A[r][j];
      b[i] -= z * b[r];
    }
    col.pb(c);
    r++;
  }
  rep(i, r, n) if(abs(b[i]) > eps) return {-1, {}};
  vd x(m, numeric_limits<double>::quiet_NaN());
  vector<bool>pivot(m);
  for(int c : col) pivot[c] = 1;
  rep(i, 0, r){
    bool unique = 1;
    rep(c, 0, m) if(!pivot[c] and abs(A[i][c]) > eps) unique =
        0;
    if(unique) x[col[i]] = b[i];
  }
  return {r, x};
}
```

SolveLinearBinary.h

Description: Solves A*x=b over F2. Returns rank, x, or -1, .
Time: $\mathcal{O}\left(N^2M\right)$

353e56, 29 lines

```
typedef bitset<1000>bs;

pair<int, bs> solveLinear(vector<bs>A, vi b, int m){
  int n = sz(A), rank = 0, br;
  assert(m <= sz(bs()));
  vi col(m);
  iota(all(col), 0);
  rep(i, 0, n){
    for(br = i; br < n; br++) if(A[br].any()) break;
    if(br == n){
      rep(j, i, n) if(b[j]) return {-1, {}};
      break;
    }
    int bc = A[br]._Find_next(i - 1);
    swap(A[i], A[br]);
    swap(b[i], b[br]);
    swap(col[i], col[bc]);
    rep(j, 0, n) if(A[j][i] != A[j][bc]) A[j].flip(i), A[j].
        flip(bc);
    rep(j, i + 1, n) if(A[j][i]) b[j] ^= b[i], A[j] ^= A[i];
    rank++;
  }
  bs x;
  for(int i = rank; i--;){
    if(!b[i]) continue;
    x[col[i]] = 1;
    rep(j, 0, i) b[j] ^= A[j][i];
  }
  return {rank, x};
}
```

MatrixInverse.h

Description: Invert matrix A. Returns rank; result is stored in A unless singular (rank < n). Can easily be extended to prime moduli; for prime powers, repeatedly set $A^{-1} = A^{-1}(2I - AA^{-1})$ (mod p^k) where A^{-1} starts as the inverse of A mod p, and k is doubled in each step.
Time: $\mathcal{O}\left(n^3\right)$

ebfff6, 35 lines

```
int matInv(vector<vector<double>>& A){
  int n = sz(A); vi col(n);
  vector<vector<double>> tmp(n, vector<double>(n));
  rep(i, 0, n) tmp[i][i] = 1, col[i] = i;

  rep(i, 0, n){
    int r = i, c = i;
    rep(j, i, n) rep(k, i, n)
      if(fabs(A[j][k]) > fabs(A[r][c]))
        r = j, c = k;
    if(fabs(A[r][c]) < 1e-12) return i;
    A[i].swap(A[r]); tmp[i].swap(tmp[r]);
    rep(j, 0, n)
      swap(A[j][i], A[j][c]), swap(tmp[j][i], tmp[j][c]);
    swap(col[i], col[c]);
    double v = A[i][i];
    rep(j, i+1, n){
      double f = A[j][i] / v;
      A[j][i] = 0;
      rep(k, i+1, n) A[j][k] -= f*A[i][k];
      rep(k, 0, n) tmp[j][k] -= f*tmp[i][k];
    }
    rep(j, i+1, n) A[i][j] /= v;
    rep(j, 0, n) tmp[i][j] /= v;
    A[i][i] = 1;
  }

  for(int i = n-1; i > 0; --i) rep(j, 0, i){
    double v = A[j][i];
    rep(k, 0, n) tmp[j][k] -= v*tmp[i][k];
  }

  rep(i, 0, n) rep(j, 0, n) A[col[i]][col[j]] = tmp[i][j];
  return n;
}
```

4.4 Fourier transforms

FastFourierTransform.h

Description: fft(a) computes $\hat{f}(k) = \sum_x a[x] \exp(2\pi i \cdot kx/N)$ for all k. N must be a power of 2. Useful for convolution: `conv(a, b) = c`, where $c[x] = \sum a[i]b[x-i]$. For convolution of complex numbers or more than two vectors: FFT, multiply pointwise, divide by n, reverse(start+1, end), FFT back. Rounding is safe if $(\sum a_i^2 + \sum b_i^2)\log_2 N < 9 \cdot 10^{14}$ (in practice 10^{16}; higher for random inputs). Otherwise, use NTT/FFTMod.
Time: $\mathcal{O}\left(N \log N\right)$ with $N = |A| + |B|$ ($\sim 1s$ for $N = 2^{22}$)

00ced6, 35 lines

```
typedef complex<double> C;
typedef vector<double> vd;
void fft(vector<C>& a){
  int n = sz(a), L = 31 - __builtin_clz(n);
  static vector<complex<long double>> R(2, 1);
  static vector<C> rt(2, 1);  // (^ 10% faster if double)
  for(static int k = 2; k < n; k *= 2){
    R.resize(n); rt.resize(n);
    auto x = polar(1.0L, acos(-1.0L) / k);
    rep(i, k, 2*k) rt[i] = R[i] = i&1 ? R[i/2] * x : R[i/2];
  }
  vi rev(n);
  rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
  rep(i, 0, n) if(i < rev[i]) swap(a[i], a[rev[i]]);
  for(int k = 1; k < n; k *= 2)
    for(int i = 0; i < n; i += 2 * k) rep(j, 0, k){
      C z = rt[j+k] * a[i+j+k]; // (25% faster if hand-rolled)
      a[i + j + k] = a[i + j] - z;
      a[i + j] += z;
    }
}
vd conv(const vd& a, const vd& b){
  if(a.empty() || b.empty()) return {};
  vd res(sz(a) + sz(b) - 1);
  int L = 32 - __builtin_clz(sz(res)), n = 1 << L;
  vector<C> in(n), out(n);
  copy(all(a), begin(in));
  rep(i, 0, sz(b)) in[i].imag(b[i]);
  fft(in);
  for(C& x : in) x *= x;
  rep(i, 0, n) out[i] = in[-i & (n - 1)] - conj(in[i]);
  fft(out);
  rep(i, 0, sz(res)) res[i] = imag(out[i]) / (4 * n);
  return res;
}
```

FastFourierTransformMod.h

Description: Exact convolution modulo M using three NTT primes and CRT. Supports transform lengths up to 2^{21}.
Time: $\mathcal{O}\left(NlogN\right)$

d2f979, 60 lines

```
typedef vector<ll> vl;
namespace conv_mod {
template<int P> ll pw(ll a, ll n){
  ll r = 1;
  for(; n; n >>= 1, a = a*a%P) if(n&1) r = r*a%P;
  return r;
}
template<int P, int G> void ntt(vl& a, bool inv){
  int n = sz(a);
  for(int i = 1, j = 0; i < n; i++){
    int bit = n >> 1;
    for(; j&bit; bit >>= 1) j ^= bit;
    j ^= bit;
    if(i < j) swap(a[i], a[j]);
  }
  for(int len = 2; len <= n; len <<= 1){
    ll wl = pw<P>(G, (P-1)/len);
    if(inv) wl = pw<P>(wl, P-2);
    vl W(len/2, 1);
    rep(j, 1, len/2) W[j] = W[j-1]*wl%P;
    for(int i = 0; i < n; i += len) rep(j, 0, len/2){
      ll u = a[i+j], v = a[i+j+len/2]*W[j]%P;
      a[i+j] = u+v < P ? u+v : u+v-P;
      a[i+j+len/2] = u-v < 0 ? u-v+P : u-v;
    }
  }
  if(inv){
    ll ni = pw<P>(n, P-2);
    for(ll &x : a) x = x*ni%P;
  }
}
template<int P, int G> vl conv(const vl& a, const vl& b, int n,
      int s){
  vl x(n), y(n);
  rep(i, 0, sz(a)) x[i] = a[i]%P;
  rep(i, 0, sz(b)) y[i] = b[i]%P;
  ntt<P, G>(x, 0); ntt<P, G>(y, 0);
  rep(i, 0, n) x[i] = x[i]*y[i]%P;
  ntt<P, G>(x, 1); x.resize(s);
  return x;
}
}

template<int M> vl convMod(const vl& a, const vl& b){
  if(a.empty() || b.empty()) return {};
  const ll p0 = 998244353, p1 = 1004535809, p2 = 469762049;
  int s = sz(a)+sz(b)-1, n = 1;
  while(n < s) n <<= 1;
  auto x = conv_mod::conv<p0, 3>(a, b, n, s);
  auto y = conv_mod::conv<p1, 3>(a, b, n, s);
  auto z = conv_mod::conv<p2, 3>(a, b, n, s);
  ll i01 = conv_mod::pw<p1>(p0, p1-2);
  ll p01 = p0*p1, i012 = conv_mod::pw<p2>(p01%p2, p2-2);
```

```
  vl ret(s);
  rep(i, 0, s){
    ll b1 = (y[i]-x[i]+p1)%p1*i01%p1;
    ll b2 = (z[i]-(x[i]+p0*b1)%p2+p2)%p2*i012%p2;
    ret[i] = (x[i]+(__int128)p0*b1+(__int128)p01*b2)%M;
  }
  return ret;
}
```

NumberTheoreticTransform.h

Description: NTT/convolution mod 998244353; inputs in $[0, mod)$, size up to 2^{23}.

Time: $\mathcal{O}(NlogN)$

15d138, 39 lines

```
const int N = (1<<20); // change if necessary, has to be 2^k
const ll mod = 998244353, G = 3;
ll mul(ll a, ll b){ return a * b % mod; }
ll add(ll a, ll b){ return a + b < mod ? a + b : a + b - mod; }
ll sub(ll a, ll b){ return a < b ? a - b + mod : a - b; }
ll Pow(ll a, ll e){ ll r = 1; for(; e; e >>= 1, a = mul(a, a))
    if(e & 1) r = mul(r, a); return r; }

struct NTT {
  ll w[N];
  NTT() { // w[k + j] = (2k-th root)^j
    for (int k = 1; k < N; k <<= 1) {
      ll dw = Pow(G, (mod - 1) / (2 * k)); w[k] = 1;
      for (int j = 1; j < k; ++j) w[k + j] = mul(w[k + j - 1],
          dw);
    }
  } // 0 <= a[i] < P
  void operator()(vector<ll>& a, bool inv = false) {
    int n = sz(a);
    for (int j = 1, x = 0; j < n - 1; ++j) {
      for (int k = n >> 1; (x ^= k) < k; k >>= 1);
      if (j < x) swap(a[x], a[j]);
    }
    for (int L = 2; L <= n; L <<= 1) {
      int dl = L >> 1;
      for (int i = 0; i < n; i += L) {
        for (int j = i, x = dl; j < i + dl; ++j, ++x) {
          ll tmp = mul(a[j + dl], w[x]);
          a[j + dl] = sub(a[j], tmp);
          a[j] = add(a[j], tmp);
        }
      }
    }
    if (inv) {
      reverse(1 + all(a));
      ll invn = Pow(n, mod - 2);
      for (int i = 0; i < n; ++i)
        a[i] = mul(a[i], invn);
    }
  }
} ntt;
```

FastSubsetTransform.h

Description: Transform to a basis with fast convolutions of the form $c[z] = \sum_{z=x \oplus y} a[x] \cdot b[y]$, where $\oplus$ is one of AND, OR, XOR. The size of a must be a power of two.

Time: $\mathcal{O}(N \log N)$

bf4cb5, 20 lines

```
void FST(vi& a, bool inv, int mod = 0){
  for(int n = sz(a), step = 1; step < n; step *= 2){
    for(int i = 0; i < n; i += 2 * step) rep(j, i, i+step){
      int &u = a[j], &v = a[j + step], x = u, y = v;
      if(mod){
        if(inv){ u = y - x; if(u < 0) u += mod; v = x; }
        else { u = y; v = x + y; if(v >= mod) v -= mod; }
      }
      else tie(u, v) = inv ? pii(y - x, x) : pii(y, x + y); //
          AND
        inv ? pii(v, u - v) : pii(u + v, u); // OR
        pii(u + v, u - v);                   // XOR
    }
  }
  if(inv) for(int& x : a) x /= sz(a); // XOR only; mod: *
      modpow(sz(a), mod-2)
}
vi conv(vi a, vi b, int mod = 0){
  FST(a, 0, mod); FST(b, 0, mod);
  rep(i, 0, sz(a)) a[i] = mod ? (int)((ll)a[i]*b[i] % mod) : a[
      i]*b[i];
  FST(a, 1, mod); return a;
}
```

Number theory (5)

5.1 Modular arithmetic

5.1.1 Nice Primes

For NTT,
$m = 9223372036737335297 = 549755813881 \times 2^{24} + 1, g = 3$ (very close to long long limit $2^{63} - 1$) For hashing we can also use $MOD = 1000000000000000003, 1000000000000000033, BASE = 911382323$

ModularArithmetic.h

Description: Operators for modular arithmetic. You need to set mod to some number first and then you can use the structure.

"euclid.h" ea1ac7, 19 lines

```
const ll mod = 17; // change to something else
struct Mod {
  ll x;
  Mod(ll y) : Mod(y % mod + mod, 0){}
  Mod(ll y, int) : x(y < mod ? y : y - mod){}
  Mod operator+(Mod b){ return {x + b.x, 0}; }
  Mod operator-(Mod b){ return {x - b.x + mod, 0}; }
  Mod operator*(Mod b){ return {x * b.x % mod, 0}; }
  Mod operator/(Mod b){ return *this * invert(b); }
  Mod invert(Mod a){
    ll x, y, g = euclid(a.x, mod, x, y);
    assert(g == 1); return x;
  }
  Mod operator^(ll e){
    if(!e) return 1;
    Mod r = *this ^ (e / 2); r = r * r;
    return e&1 ? *this * r : r;
  }
};
```

ModInverse.h

Description: Inverses of [1, n) modulo a prime, assuming n <= mod.

Time: $\mathcal{O}(n)$

c4afb5, 6 lines

```
vector<ll> inverses(int n, ll mod){
  vector<ll> inv(n);
  if(n > 1) inv[1] = 1;
  rep(i, 2, n) inv[i] = mod - mod / i * inv[mod % i] % mod;
  return inv;
}
```

ModPow.h

1e790e, 9 lines

```
const ll mod = 1000000007; // faster if const

ll modpow(ll b, ll e, ll m = mod){
  ll ans = 1;
  b %= m;
  for(; e; b = b * b % m, e /= 2)
    if(e & 1) ans = ans * b % m;
  return ans;
}
```

ModLog.h

Description: Returns the smallest $x > 0$ s.t. $a^x = b \pmod m$, or -1 if no such x exists. modLog(a, 1, m) can be used to calculate the order of a.

Time: $\mathcal{O}\left(\sqrt{m}\right)$

c040b8, 11 lines

```
ll modLog(ll a, ll b, ll m){
  ll n = (ll) sqrt(m) + 1, e = 1, f = 1, j = 1;
  unordered_map<ll, ll> A;
  while(j <= n && (e = f = e * a % m) != b % m)
    A[e * b % m] = j++;
  if(e == b % m) return j;
  if(__gcd(m, e) == __gcd(m, b))
    rep(i, 2, n+2) if(A.count(e = e * f % m))
      return n * i - A[e];
  return -1;
}
```

ModSum.h

Description: Sums of mod'ed arithmetic progressions.
modsum(to, c, k, m) $= \sum_{i=0}^{to-1} (ki + c)\%m$. divsum is similar but for floored division.

Time: $\log(m)$, with a large constant.

5c5bc5, 16 lines

```
typedef unsigned long long ull;
ull sumsq(ull to){ return to / 2 * ((to-1) | 1); }

ull divsum(ull to, ull c, ull k, ull m){
  ull res = k / m * sumsq(to) + c / m * to;
  k %= m; c %= m;
  if(!k) return res;
  ull to2 = (to * k + c) / m;
  return res + (to - 1) * to2 - divsum(to2, m-1 - c, m, k);
}

ll modsum(ull to, ll c, ll k, ll m){
  c = ((c % m) + m) % m;
  k = ((k % m) + m) % m;
  return to * c + k * sumsq(to) - m * divsum(to, c, k, m);
}
```

ModMulLL.h

Description: Portable a*b mod M for $0 \le a, b < M < 2^{64}$.

Time: $\mathcal{O}(1)$ for modmul, $\mathcal{O}(loge)$ for modpow

5cbaa6, 10 lines

```
typedef unsigned long long ull;
ull modmul(ull a, ull b, ull M){
  return (ull)((unsigned __int128)a * b % M);
}
ull modpow(ull a, ull e, ull M){
  ull r = 1;
  for(; e; e >>= 1, a = modmul(a, a, M))
    if(e & 1) r = modmul(r, a, M);
  return r;
}
```

ModSqrt.h

Description: Tonelli-Shanks algorithm for modular square roots. Finds x s.t. $x^2 = a \pmod p$ ($-x$ gives the other solution).
Time: $\mathcal{O}\left(\log^2 p\right)$ worst case, $\mathcal{O}\left(\log p\right)$ for most p

"ModPow.h" 19a793, 24 lines

```
ll sqrt(ll a, ll p){
  a %= p; if(a < 0) a += p;
  if(a == 0) return 0;
  assert(modpow(a, (p-1)/2, p) == 1); // else no solution
  if(p % 4 == 3) return modpow(a, (p+1)/4, p);
  // a^(n+3)/8 or 2^(n+3)/8 * 2^(n-1)/4 works if p % 8 == 5
  ll s = p - 1, n = 2;
  int r = 0, m;
  while(s % 2 == 0)
    ++r, s /= 2;
  while(modpow(n, (p - 1) / 2, p) != p - 1) ++n;
  ll x = modpow(a, (s + 1) / 2, p);
  ll b = modpow(a, s, p), g = modpow(n, s, p);
  for(;; r = m){
    ll t = b;
    for(m = 0; m < r && t != 1; ++m)
      t = t * t % p;
    if(m == 0) return x;
    ll gs = modpow(g, 1LL << (r - m - 1), p);
    g = gs * gs % p;
    x = x * gs % p;
    b = b * g % p;
  }
}
```

5.2 Primality

FastEratosthenes.h

Description: Prime sieve for generating all primes smaller than LIM.
Time: LIM=1e9 $\approx$ 1.5s

f9d471, 20 lines

```
const int LIM = 1e6;
bitset<LIM> isPrime;
vi eratosthenes(){
  const int S = (int)round(sqrt(LIM)), R = LIM / 2;
  vi pr = {2}, sieve(S+1); pr.reserve((size_t)(LIM/log(LIM)
      *1.1));
  vector<pii> cp;
  for(int i = 3; i <= S; i += 2) if(!sieve[i]){
    cp.push_back({i, i * i / 2});
    for(int j = i * i; j <= S; j += 2 * i) sieve[j] = 1;
  }
  for(int L = 1; L <= R; L += S){
    array<bool, S> block{};
    for(auto &[p, idx] : cp)
      for(int i=idx; i < S+L; idx = (i+=p)) block[i-L] = 1;
    rep(i, 0, min(S, R - L))
      if(!block[i]) pr.push_back((L + i) * 2 + 1);
  }
  for(int i : pr) isPrime[i] = 1;
  return pr;
}
```

PrimeCount.h

Description: Number of primes $\le N$ (Lucy-style). countPrimes(1e13) = 346065536839.
Time: $\mathcal{O}\left(N^{3/4}/\log N\right)$, about 0.03s for $N = 10^{11}$

cc786b, 19 lines

```
ll countPrimes(ll N){
  if(N <= 1) return 0;
  int sq = (int)sqrtl(N), pc = 0;
  vector<ll> big((sq + 1) / 2), small(sq + 1);
  rep(i, 1, sq + 1) small[i] = (i - 1) / 2;
  rep(i, 0, sz(big)) big[i] = (N / (2 * i + 1) - 1) / 2;
  vector<bool> skip(sq + 1);
  for(int p = 3; p <= sq; p += 2) if(!skip[p]){
    for(int j = p; j <= sq; j += 2 * p) skip[j] = 1;
    rep(j, 0, min((ll)sz(big), (N / p / p + 1) / 2)){
      ll prod = (ll)(2 * j + 1) * p;
      big[j] -= (prod > sq ? small[(ll)((double)N / prod)] :
          big[prod / 2]) - pc;
    }
    for(int j = sq, q = sq / p; q >= p; --q) for(; j >= q * p;
        --j)
      small[j] -= small[q] - pc;
    ++pc;
  }
  return big[0] + 1;
}
```

MillerRabin.h

Description: Deterministic Miller-Rabin primality test. Guaranteed to work for numbers up to $7 \cdot 10^{18}$; for larger numbers, use Python and extend A randomly.
Time: 7 times the complexity of $a^b \mod c$.

"ModMulLL.h" 60dcd1, 12 lines

```
bool isPrime(ull n){
  if(n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
  ull A[] = {2, 325, 9375, 28178, 450775, 9780504, 1795265022},
      s = __builtin_ctzll(n-1), d = n >> s;
  for(ull a : A){   // ^ count trailing zeroes
    ull p = modpow(a%n, d, n), i = s;
    while(p != 1 && p != n - 1 && a % n && i--)
      p = modmul(p, p, n);
    if(p != n-1 && i != s) return 0;
  }
  return 1;
}
```

Factor.h

Description: Pollard-rho randomized factorization algorithm. Returns prime factors of a number, in arbitrary order (e.g. 2299 -> {11, 19, 11}).
Time: $\mathcal{O}\left(n^{1/4}\right)$, less for numbers with small factors.

"ModMulLL.h", "MillerRabin.h" d8d98d, 18 lines

```
ull pollard(ull n){
  ull x = 0, y = 0, t = 30, prd = 2, i = 1, q;
  auto f = [&](ull x){ return modmul(x, x, n) + i; };
  while(t++ % 40 || __gcd(prd, n) == 1){
    if(x == y) x = ++i, y = f(x);
    if((q = modmul(prd, max(x, y) - min(x, y), n))) prd = q;
    x = f(x), y = f(f(y));
  }
  return __gcd(prd, n);
}
vector<ull> factor(ull n){
  if(n == 1) return {};
  if(isPrime(n)) return {n};
  ull x = pollard(n);
  auto l = factor(x), r = factor(n / x);
  l.insert(l.end(), all(r));
  return l;
}
```

PrimitiveRoot.h

Description: Smallest primitive root of n, or 0 if none exists (they exist only for $n = 1, 2, 4, p^k, 2p^k$).
Time: Pollard-rho on n and $\phi(n)$, plus a few modpow per candidate

"Factor.h" d2d027, 16 lines

```
ull primitiveRoot(ull n){
  if(n <= 4) return n - 1;
  auto f = factor(n); sort(all(f));
  int c2 = count(all(f), 2ULL);
  f.erase(unique(all(f)), f.end());
  if(c2 > 1 || sz(f) - (c2 > 0) != 1) return 0;
  ull ph = n;
  for(ull p : f) ph = ph / p * (p - 1);
  auto q = factor(ph); sort(all(q));
  q.erase(unique(all(q)), q.end());
  for(ull g = 2;; g++) if(__gcd(g, n) == 1){
    bool ok = 1;
    for(ull p : q) ok &= modpow(g, ph / p, n) != 1;
    if(ok) return g;
  }
}
```

5.3 Divisibility

euclid.h

Description: Finds two integers x and y, such that $ax + by = \gcd(a, b)$. If you just need gcd, use the built in __gcd instead. If a and b are coprime, then x is the inverse of $a \pmod b$.

33ba8f, 5 lines

```
ll euclid(ll a, ll b, ll &x, ll &y){
  if(!b) return x = 1, y = 0, a;
  ll d = euclid(b, a % b, y, x);
  return y -= a/b * x, d;
}
```

CRT.h

Description: Chinese Remainder Theorem.
crt(a, m, b, n) computes x such that $x \equiv a \pmod m$, $x \equiv b \pmod n$. If $|a| < m$ and $|b| < n$, x will obey $0 \le x < \text{lcm}(m, n)$. Assumes $mn < 2^{62}$.
Time: $\log(n)$

"euclid.h" 04d93a, 7 lines

```
ll crt(ll a, ll m, ll b, ll n){
  if(n > m) swap(a, b), swap(m, n);
  ll x, y, g = euclid(m, n, x, y);
  assert((a - b) % g == 0); // else no solution
  x = (b - a) % n * x % n / g * m + a;
  return x < 0 ? x + m*n/g : x;
}
```

5.3.1 Bézout's identity

For $a \neq, b \neq 0$, then $d = gcd(a, b)$ is the smallest positive integer for which there are integer solutions to

$$ax + by = d$$

If (x, y) is one solution, then all solutions are given by

$$\left(x + \frac{kb}{\gcd(a,b)}, y - \frac{ka}{\gcd(a,b)}\right), \quad k \in \mathbb{Z}$$

phiFunction.h

Description: *Euler's* ϕ function is defined as $\phi(n) :=$ # of positive integers $\le n$ that are coprime with n. $\phi(1) = 1$, p prime $\Rightarrow \phi(p^k) = (p-1)p^{k-1}$, m, n coprime $\Rightarrow \phi(mn) = \phi(m)\phi(n)$. If $n = p_1^{k_1} p_2^{k_2} ... p_r^{k_r}$ then $\phi(n) = (p_1 - 1)p_1^{k_1-1} ... (p_r - 1)p_r^{k_r-1}$. $\phi(n) = n \cdot \prod_{p|n}(1 - 1/p)$.
$\sum_{d|n} \phi(d) = n$, $\sum_{1 \le k \le n, \gcd(k,n)=1} k = n\phi(n)/2, n > 1$
Euler's thm: a, n coprime $\Rightarrow a^{\phi(n)} \equiv 1 \pmod n$.
Fermat's little thm: p prime $\Rightarrow a^{p-1} \equiv 1 \pmod p$ $\forall a$.

cf7d6d, 8 lines

```
const int LIM = 5000000;
int phi[LIM];

void calculatePhi(){
```

```
  rep(i, 0, LIM) phi[i] = i&1 ? i : i/2;
  for(int i = 3; i < LIM; i += 2) if(phi[i] == i)
    for(int j = i; j < LIM; j += i) phi[j] -= phi[j] / i;
}
```

5.4 Fractions

ContinuedFractions.h

Description: Given N and a real number $x \geq 0$, finds the closest rational approximation p/q with $p, q \leq N$. It will obey $|p/q - x| \leq 1/qN$.
For consecutive convergents, $p_{k+1}q_k - q_{k+1}p_k = (-1)^k$. ($p_k/q_k$ alternates between $> x$ and $< x$.) If x is rational, y eventually becomes ∞; if x is the root of a degree 2 polynomial the a's eventually become cyclic.
Time: $\mathcal{O}(\log N)$

dd6c5e, 21 lines

```
typedef double d; // for N ~ 1e7; long double for N ~ 1e9
pair<ll, ll> approximate(d x, ll N){
  ll LP = 0, LQ = 1, P = 1, Q = 0, inf = LLONG_MAX; d y = x;
  for(;;){
    ll lim = min(P ? (N-LP) / P : inf, Q ? (N-LQ) / Q : inf),
       a = (ll)floor(y), b = min(a, lim),
       NP = b*P + LP, NQ = b*Q + LQ;
    if(a > b){
      // If b > a/2, we have a semi-convergent that gives us a
      // better approximation; if b = a/2, we *may* have one.
      // Return {P, Q} here for a more canonical approximation.
      return (abs(x - (d)NP / (d)NQ) < abs(x - (d)P / (d)Q)) ?
        make_pair(NP, NQ) : make_pair(P, Q);
    }
    if(abs(y = 1/(y - (d)a)) > 3*N){
      return {NP, NQ};
    }
    LP = P; P = NP;
    LQ = Q; Q = NQ;
  }
}
```

FracBinarySearch.h

Description: Given f and N, finds the smallest fraction $p/q \in [0, 1]$ such that $f(p/q)$ is true, and $p, q \leq N$. You may want to throw an exception from f if it finds an exact solution, in which case N can be removed.
Usage: `fracBS([](Frac f){ return f.p>=3*f.q; }, 10); // {1, 3}`
Time: $\mathcal{O}(\log(N))$

27ab3e, 25 lines

```
struct Frac { ll p, q; };

template<class F>
Frac fracBS(F f, ll N){
  bool dir = 1, A = 1, B = 1;
  Frac lo{0, 1}, hi{1, 1}; // Set hi to 1/0 to search (0, N]
  if(f(lo)) return lo;
  assert(f(hi));
  while(A || B){
    ll adv = 0, step = 1; // move hi if dir, else lo
    for(int si = 0; step; (step *= 2) >>= si){
      adv += step;
      Frac mid{lo.p * adv + hi.p, lo.q * adv + hi.q};
      if(abs(mid.p) > N || mid.q > N || dir == !f(mid)){
        adv -= step; si = 2;
      }
    }
    hi.p += lo.p * adv;
    hi.q += lo.q * adv;
    dir = !dir;
    swap(lo, hi);
    A = B; B = !!adv;
  }
  return dir ? hi : lo;
}
```

5.5 Pythagorean Triples

The Pythagorean triples are uniquely generated by

$$a = k \cdot (m^2 - n^2),\ \ b = k \cdot (2mn),\ \ c = k \cdot (m^2 + n^2),$$

with $m > n > 0$, $k > 0$, $m \perp n$, and either m or n even.

5.6 Primes

$p = 962592769$ is such that $2^{21} \mid p - 1$, which may be useful. For hashing use 970592641 (31-bit number), 31443539979727 (45-bit), 3006703054056749 (52-bit). There are 78498 primes less than 1 000 000.

Primitive roots exist modulo any prime power p^a, except for $p = 2, a > 2$, and there are $\phi(\phi(p^a))$ many. For $p = 2, a > 2$, the group $\mathbb{Z}_{2^a}^\times$ is instead isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_{2^{a-2}}$.

5.7 Estimates

$\sum_{d|n} d = O(n \log \log n)$.

The number of divisors of n is at most around 100 for $n < 5e4$, 500 for $n < 1e7$, 2000 for $n < 1e10$, 200 000 for $n < 1e19$.

5.8 Mobius Function

$$\mu(n) = \begin{cases} 0 & n \text{ is not square free} \\ 1 & n \text{ has even number of prime factors} \\ -1 & n \text{ has odd number of prime factors} \end{cases}$$

Mobius Inversion:

$$g(n) = \sum_{d|n} f(d) \Leftrightarrow f(n) = \sum_{d|n} \mu(d) g(n/d)$$

Other useful formulas/forms:

$\sum_{d|n} \mu(d) = [n = 1]$ (very useful)

$g(n) = \sum_{n|d} f(d) \Leftrightarrow f(n) = \sum_{n|d} \mu(d/n) g(d)$

$g(n) = \sum_{1 \leq m \leq n} f(\lfloor \frac{n}{m} \rfloor) \Leftrightarrow f(n) = \sum_{1 \leq m \leq n} \mu(m) g(\lfloor \frac{n}{m} \rfloor)$

Combinatorial (6)

6.1 Permutations

6.1.1 Factorial

n	1	2	3	4	5	6	7	8	9	10
$n!$	1	2	6	24	120	720	5040	40320	362880	3628800
n	11	12	13	14	15	16	17			
$n!$	4.0e7	4.8e8	6.2e9	8.7e10	1.3e12	2.1e13	3.6e14			
n	20	25	30	40	50	100	150	171		
$n!$	2e18	2e25	3e32	8e47	3e64	9e157	6e262	>DBL_MAX		

IntPerm.h

Description: Permutation -> integer conversion. (Not order preserving.) Integer -> permutation can use a lookup table.
Time: $\mathcal{O}(n)$

044568, 6 lines

```
int permToInt(vi& v){
  int use = 0, i = 0, r = 0;
  for(int x:v) r = r * ++i + __builtin_popcount(use & -(1<<x)),
    use |= 1 << x;                    // (note: minus, not ~!)
  return r;
}
```

6.1.2 Cycles

Let $g_S(n)$ be the number of n-permutations whose cycle lengths all belong to the set S. Then

$$\sum_{n=0}^{\infty} g_S(n) \frac{x^n}{n!} = \exp\left(\sum_{n \in S} \frac{x^n}{n}\right)$$

6.1.3 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1) + D(n-2)) = nD(n-1) + (-1)^n = \left\lfloor \frac{n!}{e} \right\rceil$$

6.1.4 Burnside's lemma

Given a group G of symmetries and a set X, the number of elements of X *up to symmetry* equals

$$\frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts "configurations" (of some sort) of length n, we can ignore rotational symmetry using $G = \mathbb{Z}_n$ to get

$$g(n) = \frac{1}{n} \sum_{k=0}^{n-1} f(\gcd(n, k)) = \frac{1}{n} \sum_{k|n} f(k) \phi(n/k).$$

6.2 Partitions and subsets

6.2.1 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1,\ p(n) = \sum_{k \in \mathbb{Z} \setminus \{0\}} (-1)^{k+1} p(n - k(3k-1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	~2e5	~2e8

6.2.2 Lucas' Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + ... + n_1 p + n_0$ and $m = m_k p^k + ... + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^{k} \binom{n_i}{m_i} \pmod{p}$.

6.2.3 Binomials

multinomial.h

Description: Computes $\binom{k_1+\cdots+k_n}{k_1,k_2,\ldots,k_n} = \frac{(\sum k_i)!}{k_1!k_2!\ldots k_n!}$. a0a312, 5 lines

```
ll multinomial(vi& v){
  ll c = 1, m = v.empty() ? 1 : v[0];
  rep(i, 1, sz(v)) rep(j, 0, v[i]) c = c * ++m / (j+1);
  return c;
}
```

6.3 General purpose numbers

6.3.1 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t-1}$ (FFT-able).
$B[0,\ldots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \ldots]$

Sums of powers:

$$\sum_{i=1}^{n} n^m = \frac{1}{m+1}\sum_{k=0}^{m}\binom{m+1}{k}B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^{\infty} f(i) = \int_m^\infty f(x)dx - \sum_{k=1}^{\infty}\frac{B_k}{k!}f^{(k-1)}(m)$$
$$\approx \int_m^\infty f(x)dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

6.3.2 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),\ c(0,0) = 1$$
$$\textstyle\sum_{k=0}^{n} c(n,k)x^k = x(x+1)\ldots(x+n-1)$$

$c(8,k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$
$c(n,2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \ldots$

6.3.3 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j:s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j:s s.t. $\pi(j) \geq j$, k j:s s.t. $\pi(j) > j$.

$$E(n,k) = (n-k)E(n-1,k-1) + (k+1)E(n-1,k)$$
$$E(n,0) = E(n,n-1) = 1$$
$$E(n,k) = \sum_{j=0}^{k}(-1)^j\binom{n+1}{j}(k+1-j)^n$$

6.3.4 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$
$$S(n,1) = S(n,n) = 1$$
$$S(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}j^n$$

6.3.5 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \ldots.$ For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod p$$

6.3.6 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i: $n_1 n_2 \cdots n_k n^{k-2}$
with degrees d_i: $(n-2)!/((d_1-1)!\cdots(d_n-1)!)$

6.3.7 Catalan numbers

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$
$$C_0 = 1,\ C_{n+1} = \frac{2(2n+1)}{n+2}C_n,\ C_{n+1} = \sum C_i C_{n-i}$$

$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \ldots$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

Graph (7)

7.1 Fundamentals

BellmanFord.h

Description: Single-source shortest paths with negative cycles. Unreachable vertices have INF; vertices after a negative cycle have -INF.
Time: $\mathcal{O}(VE)$ 470aae, 28 lines

```
struct BellmanFord{
  struct Edge{ int u, v; ll w; };
  const ll INF = 1LL << 62;
  int n;
  vector<Edge>ed;
  BellmanFord(int N): n(N){}
  void addEdge(int u, int v, ll w){
    ed.pb({u, v, w});
  }
  pair<vector<ll>, vi> calc(int s){
    vector<ll>d(n, INF);
    vi par(n, -1);
    d[s] = 0;
    for(int i = 0, ch = 1; i < n && ch; i++){
      ch = 0;
      for(auto [u, v, w] : ed) if(d[u] != INF){
        ll nd = max(-INF, d[u] + w);
        if(nd < d[v]) d[v] = i == n - 1 ? -INF : nd, par[v] = u
            , ch = 1;
      }
    }
    for(int i = 0, ch = 1; i < n && ch; i++){
      ch = 0;
      for(auto [u, v, w] : ed)
        if(d[u] == -INF && d[v] != -INF) d[v] = -INF, ch = 1;
    }
    return {d, par};
  }
};
```

FloydWarshall.h

Description: Calculates all-pairs shortest path in a directed graph that might have negative edge weights. Input is an distance matrix m, where $m[i][j] =$ `inf` if i and j are not adjacent. As output, $m[i][j]$ is set to the shortest distance between i and j, `inf` if no path, or `-inf` if the path goes through a negative-weight cycle.
Time: $\mathcal{O}(N^3)$ 531245, 14 lines

```
const ll inf = 1LL << 62;
void floydWarshall(vector<vector<ll>>& m){
  int n = sz(m);
  rep(i, 0, n) m[i][i] = min(m[i][i], 0LL);
  rep(k, 0, n) rep(i, 0, n) rep(j, 0, n)
    if(m[i][k] != inf && m[k][j] != inf){
      auto newDist = max(m[i][k] + m[k][j], -inf);
      m[i][j] = min(m[i][j], newDist);
    }
  rep(k, 0, n) if(m[k][k] < 0) rep(i, 0, n) rep(j, 0, n)
    if(m[i][k] != inf && m[k][j] != inf) m[i][j] = -inf;
}
// No negative edges, inf = 1e18: branch-free, vectorizes, ~2x
    faster:
// rep(k, 0, n) rep(i, 0, n) rep(j, 0, n) m[i][j] = min(m[i][j
    ], m[i][k] + m[k][j]);
```

7.2 Network flow

PushRelabel.h

Description: Push-relabel using the highest label selection rule and the gap heuristic. Quite fast in practice. To obtain the actual flow, look at positive values only.
Time: $\mathcal{O}\left(V^2\sqrt{E}\right)$ 0ae1d4, 48 lines

```
struct PushRelabel {
  struct Edge {
    int dest, back;
    ll f, c;
  };
  vector<vector<Edge>> g;
  vector<ll> ec;
  vector<Edge*> cur;
  vector<vi> hs; vi H;
  PushRelabel(int n) : g(n), ec(n), cur(n), hs(2*n), H(n){}

  void addEdge(int s, int t, ll cap, ll rcap=0){
    if(s == t) return;
    g[s].push_back({t, sz(g[t]), 0, cap});
    g[t].push_back({s, sz(g[s])-1, 0, rcap});
  }

  void addFlow(Edge& e, ll f){
    Edge &back = g[e.dest][e.back];
    if(!ec[e.dest] && f) hs[H[e.dest]].push_back(e.dest);
    e.f += f; e.c -= f; ec[e.dest] += f;
    back.f -= f; back.c += f; ec[back.dest] -= f;
  }
  ll calc(int s, int t){
    int v = sz(g); H[s] = v; ec[t] = 1;
    vi co(2*v); co[0] = v-1;
    rep(i, 0, v) cur[i] = g[i].data();
    for(Edge& e : g[s]) addFlow(e, e.c);

    for(int hi = 0;;){
```

```
    while(hs[hi].empty()) if(!hi--) return -ec[s];
    int u = hs[hi].back(); hs[hi].pop_back();
    while(ec[u] > 0)  // discharge u
      if(cur[u] == g[u].data() + sz(g[u])){
        H[u] = 1e9;
        for(Edge& e : g[u]) if(e.c && H[u] > H[e.dest]+1)
          H[u] = H[e.dest]+1, cur[u] = &e;
        if(++co[H[u]], !--co[hi] && hi < v)
          rep(i, 0, v) if(hi < H[i] && H[i] < v)
            --co[H[i]], H[i] = v + 1;
        hi = H[u];
      } else if(cur[u]->c && H[u] == H[cur[u]->dest]+1)
        addFlow(*cur[u], min(ec[u], cur[u]->c));
      else ++cur[u];
  }
 }
 bool leftOfMinCut(int a){ return H[a] >= sz(g); }
};
```

MinCostMaxFlow.h

Description: Successive shortest path min-cost max-flow using potentials and a PBDS heap. Call setpi before maxflow if initial residual edges can have negative costs. Negative-cost cycles are not supported.
Usage: Prefer when the number of augmentations is small, or each shortest path can push
a large bottleneck flow. Usually the simplest and fastest default for ordinary MCMF.
Initial negative edges need setpi; negative cycles need the lower-bound transformation
described below, or CostScalingMCMF. PBDS requires GNU GCC/libstdc++.
Time: $\mathcal{O}(FE\log V)$, and $\mathcal{O}(VE)$ for setpi

fa77d4, 94 lines

```
const ll MCMF_INF = numeric_limits<ll>::max() / 4;
struct MCMF{
  struct edge{
    int from, to, rev;
    ll cap, cost, flow;
  };
  int N;
  vector<vector<edge>>ed;
  vi seen;
  vector<ll>dist, pi;
  vector<edge*>par;
  MCMF(int N) : N(N), ed(N), seen(N), dist(N), pi(N), par(N){}
  void addEdge(int from, int to, ll cap, ll cost){
    if(from == to) return;
    ed[from].push_back(edge{from, to, sz(ed[to]), cap, cost, 0}
        );
    ed[to].push_back(edge{to, from, sz(ed[from]) - 1, 0, -cost,
        0});
  }
  void path(int s){
    fill(all(seen), 0);
    fill(all(dist), MCMF_INF);
    fill(all(par), nullptr);
    dist[s] = 0;
    using PQ = __gnu_pbds::priority_queue<pair<ll, int>,
        greater<pair<ll, int>>>;
    PQ pq;
    vector<PQ::point_iterator> its(N);
    pq.push({0, s});
    while(!pq.empty()){
      int v = pq.top().S;
      pq.pop();
      seen[v] = 1;
      ll di = dist[v] + pi[v];
      for(edge &e : ed[v]){
        if(seen[e.to]) continue;
        if(e.cap - e.flow <= 0) continue;
        ll val = di - pi[e.to] + e.cost;
        if(val < dist[e.to]){
          dist[e.to] = val;
          par[e.to] = &e;
          if(its[e.to] == pq.end()){
            its[e.to] = pq.push({dist[e.to], e.to});
          }else{
            pq.modify(its[e.to], {dist[e.to], e.to});
          }
        }
      }
    }
    rep(i, 0, N){
      if(dist[i] != MCMF_INF){
        pi[i] = min(pi[i] + dist[i], MCMF_INF);
      }
    }
  }
  pair<ll, ll> maxflow(int s, int t){
    ll totflow = 0, totcost = 0;
    while(path(s), seen[t]){
      ll fl = MCMF_INF;
      for(edge *x = par[t]; x; x = par[x->from]){
        fl = min(fl, x->cap - x->flow);
      }
      totflow += fl;
      for(edge *x = par[t]; x; x = par[x->from]){
        x->flow += fl;
        ed[x->to][x->rev].flow -= fl;
      }
    }
    rep(i, 0, N){
      for(edge &e : ed[i]){
        totcost += e.cost * e.flow;
      }
    }
    return {totflow, totcost / 2};
  }
  // Negative edges: run setpi(s); rerun after changing source
      or flow.
  // Negative cycles: use CostScalingMCMF.
  void setpi(int s){
    fill(all(pi), MCMF_INF);
    pi[s] = 0;
    int it = N, ch = 1;
    ll v;
    while(ch-- && it--){
      rep(i, 0, N){
        if(pi[i] == MCMF_INF) continue;
        for(edge &e : ed[i]){
          if(e.cap - e.flow <= 0) continue;
          if((v = pi[i] + e.cost) < pi[e.to]){
            pi[e.to] = v;
            ch = 1;
          }
        }
      }
    }
    assert(it >= 0); // negative cost cycle
  }
};
```

MinCostMaxFlowCostScaling.h

Description: Min-cost max-flow using Dinic followed by epsilon cost-scaling push-relabel. Supports negative costs and negative-cost cycles. Costs are multiplied by N internally; make sure this and the final answer fit in ll.
Usage: Prefer for large total flow, many unit-capacity edges, integer costs, negative
edges/cycles, or min-cost circulation. SSP is usually faster for only a few augmentations.
Not suitable for floating-point costs. For pure circulation, start with any feasible flow
and run only the cost-scaling phase; do not invent s, t or run maxFlow.
Time: $\mathcal{O}\left(V^2E + VE\log(VC)\right)$, where C is the maximum absolute cost

61f08e, 147 lines

```
const ll CS_INF = numeric_limits<ll>::max() / 4;
struct CostScalingMCMF{
  struct edge{
    int from, to, rev;
    ll cap, cost, flow;
  };
  int N;
  vector<vector<edge>>ed;
  vector<ll>excess, pi;
  vi ptr, inq;
  CostScalingMCMF(int N) : N(N), ed(N), excess(N), pi(N), ptr(N
      ), inq(N){}
  // Supports negative edges/cycles with finite capacities.
  // No negative-cost self-loops or unbounded instances.
  void addEdge(int from, int to, ll cap, ll cost){
    if(from == to){
      assert(cost >= 0);
      return;
    }
    assert(-CS_INF / N <= cost && cost <= CS_INF / N);
    cost *= N;
    ed[from].push_back(edge{from, to, sz(ed[to]), cap, cost, 0}
        );
    ed[to].push_back(edge{to, from, sz(ed[from]) - 1, 0, -cost,
        0});
  }
  ll residual(edge &e){
    return e.cap - e.flow;
  }
  void push(edge &e, ll fl){
    e.flow += fl;
    ed[e.to][e.rev].flow -= fl;
    excess[e.from] -= fl;
    excess[e.to] += fl;
  }
  bool admissible(edge &e){
    return residual(e) > 0 && pi[e.from] + e.cost - pi[e.to] <
        0;
  }
  void relabel(int v, ll eps){
    ll best = -CS_INF;
    ptr[v] = -1;
    rep(i, 0, sz(ed[v])){
      edge &e = ed[v][i];
      if(residual(e) > 0 && best < pi[e.to] - e.cost){
        best = pi[e.to] - e.cost;
        ptr[v] = i;
      }
    }
    assert(ptr[v] != -1);
    pi[v] = best - eps;
  }
  void refine(ll eps){
    fill(all(excess), 0);
    fill(all(ptr), 0);
    fill(all(inq), 0);
    rep(i, 0, N){
      for(edge &e : ed[i]){
        if(admissible(e)) push(e, residual(e));
```

```
    }
  }
  queue<int> q;
  rep(i, 0, N){
    if(excess[i] > 0){
      q.push(i);
      inq[i] = 1;
    }
  }
  while(!q.empty()){
    int v = q.front();
    q.pop();
    inq[v] = 0;
    while(excess[v] > 0){
      if(ptr[v] == sz(ed[v])) relabel(v, eps);
      edge &e = ed[v][ptr[v]];
      if(!admissible(e)){
        ++ptr[v];
        continue;
      }
      ll fl = min(excess[v], residual(e));
      int to = e.to;
      push(e, fl);
      if(excess[to] > 0 && !inq[to]){
        q.push(to);
        inq[to] = 1;
      }
    }
  }
}
ll maxFlow(int s, int t){
  vi level(N), work(N);
  auto bfs = [&](){
    fill(all(level), -1);
    queue<int> q;
    level[s] = 0;
    q.push(s);
    while(!q.empty()){
      int v = q.front();
      q.pop();
      for(edge &e : ed[v]){
        if(residual(e) > 0 && level[e.to] == -1){
          level[e.to] = level[v] + 1;
          q.push(e.to);
        }
      }
    }
    return level[t] != -1;
  };
  auto dfs = [&](auto &&self, int v, ll pushed) -> ll {
    if(v == t or pushed == 0) return pushed;
    for(int &i = work[v]; i < sz(ed[v]); ++i){
      edge &e = ed[v][i];
      if(level[e.to] != level[v] + 1) continue;
      ll fl = self(self, e.to, min(pushed, residual(e)));
      if(fl == 0) continue;
      e.flow += fl;
      ed[e.to][e.rev].flow -= fl;
      return fl;
    }
    return 0;
  };
  ll flow = 0;
  while(bfs()){
    fill(all(work), 0);
    while(ll fl = dfs(dfs, s, CS_INF)) flow += fl;
  }
  return flow;
}
```

```
  pair<ll, ll> maxflow(int s, int t){
    // Find any max flow, then reduce cost by circulation.
    ll totflow = maxFlow(s, t);
    ll eps = 0;
    rep(i, 0, N){
      for(edge &e : ed[i]){
        eps = max(eps, abs(e.cost));
      }
    }
    while(eps > 0){
      refine(eps);
      if(eps == 1) break;
      eps = max(1LL, eps / 4);
    }
    __int128 totcost = 0;
    rep(i, 0, N){
      for(edge &e : ed[i]){
        totcost += (__int128)e.cost * e.flow;
      }
    }
    return {totflow, (ll)(totcost / 2 / N)};
  }
};
```

Dinic.h

Description: Flow algorithm with complexity $O(VE\log U)$ where $U = \max|\text{cap}|$. $O(\min(E^{1/2}, V^{2/3})E)$ if $U = 1$; $O(\sqrt{V}E)$ for bipartite matching, but hopcroftKarp.h is 5x faster there (yosupo bipartitematching: 0.63s vs 3.05s).

9d9667, 42 lines

```
struct Dinic{
  struct Edge{
    int to, rev;
    ll c, oc;
    ll flow(){ return max(oc - c, 0LL); } // if you need flows
  };
  vi lvl, ptr, q;
  vector<vector<Edge>>adj;
  Dinic(int n) : lvl(n), ptr(n), q(n), adj(n){}
  void addEdge(int a, int b, ll c, ll rcap = 0){
    adj[a].push_back({b, sz(adj[b]), c, c});
    adj[b].push_back({a, sz(adj[a]) - 1, rcap, rcap});
  }
  ll dfs(int v, int t, ll f){
    if(v == t || !f) return f;
    for(int &i = ptr[v]; i < sz(adj[v]); i++){
      Edge &e = adj[v][i];
      if(lvl[e.to] == lvl[v] - 1)
        if(ll p = dfs(e.to, t, min(f, e.c))){
          e.c -= p, adj[e.to][e.rev].c += p;
          return p;
        }
    }
    return 0;
  }
  ll calc(int s, int t){
    ll flow = 0; q[0] = t; // BFS from t: no dead ends in dfs
    rep(L, 0, 31) do { // 'int L=30' maybe faster for random
         data
      lvl = ptr = vi(sz(q));
      int qi = 0, qe = lvl[t] = 1;
      while(qi < qe and !lvl[s]){
        int v = q[qi++];
        for(Edge &e : adj[v])
          if(!lvl[e.to] and adj[e.to][e.rev].c >> (30 - L))
            q[qe++] = e.to, lvl[e.to] = lvl[v] + 1;
      }
      while(ll p = dfs(s, t, LLONG_MAX)) flow += p;
    }while(lvl[s]);
    return flow;
  }
  bool leftOfMinCut(int a){ return lvl[a] == 0; }
};
```

MinCut.h

Description: After running max-flow, the left side of a min-cut from s to t is given by all vertices reachable from s, only traversing edges with positive residual capacity.

GlobalMinCut.h

Description: Find a global minimum cut in an undirected graph, as represented by an adjacency matrix.
Time: $\mathcal{O}\left(V^3\right)$

8b0e19, 21 lines

```
pair<int, vi> globalMinCut(vector<vi> mat){
  pair<int, vi> best = {INT_MAX, {}};
  int n = sz(mat);
  vector<vi> co(n);
  rep(i, 0, n) co[i] = {i};
  rep(ph, 1, n){
    vi w = mat[0];
    size_t s = 0, t = 0;
    rep(it, 0, n-ph){ // O(V^2) -> O(E log V) with prio. queue
      w[t] = INT_MIN;
      s = t, t = max_element(all(w)) - w.begin();
      rep(i, 0, n) w[i] += mat[t][i];
    }
    best = min(best, {w[t] - mat[t][t], co[t]});
    co[s].insert(co[s].end(), all(co[t]));
    rep(i, 0, n) mat[s][i] += mat[t][i];
    rep(i, 0, n) mat[i][s] = mat[s][i];
    mat[0][t] = INT_MIN;
  }
  return best;
}
```

GomoryHu.h

Description: Given a list of edges representing an undirected flow graph, returns edges of the Gomory-Hu tree. The max flow between any pair of vertices is given by minimum edge weight along the Gomory-Hu tree path.
Time: $\mathcal{O}\left(V\right)$ Flow Computations

"PushRelabel.h" 0418b3, 13 lines

```
typedef array<ll, 3> Edge;
vector<Edge> gomoryHu(int N, vector<Edge> ed){
  vector<Edge> tree;
  vi par(N);
  rep(i, 1, N){
    PushRelabel D(N); // Dinic also works
    for(Edge t : ed) D.addEdge(t[0], t[1], t[2], t[2]);
    tree.push_back({i, par[i], D.calc(i, par[i])});
    rep(j, i+1, N)
      if(par[j] == par[i] && D.leftOfMinCut(j)) par[j] = i;
  }
  return tree;
}
```

7.3 Matching

hopcroftKarp.h

Description: Maximum bipartite matching. r[v] is the left match of right v.
Time: $\mathcal{O}\left(Esqrt(V)\right)$

f50330, 40 lines

```
struct HopcroftKarp{
  int n, m;
  vector<vi>g;
  vi l, r, d, q;
```

```
  HopcroftKarp(int N, int M): n(N), m(M), g(N), q(N){}
  void addEdge(int u, int v){
    g[u].pb(v);
  }
  bool dfs(int u){
    int z = exchange(d[u], 0) + 1;
    for(int v : g[u]){
      if(r[v] == -1 or (d[r[v]] == z and dfs(r[v]))){
        l[u] = v;
        r[v] = u;
        return 1;
      }
    }
    return 0;
  }
  pair<int, vi> calc(){
    int ans = 0;
    l.assign(n, -1);
    r.assign(m, -1);
    for(;;){
      int s = 0, t = 0;
      bool found = 0;
      d.assign(n, 0);
      rep(i, 0, n) if(l[i] == -1) q[t++] = i, d[i] = 1;
      while(s < t){
        int u = q[s++];
        for(int v : g[u]){
          if(r[v] == -1) found = 1;
          else if(!d[r[v]]) d[r[v]] = d[u] + 1, q[t++] = r[v];
        }
      }
      if(!found) return {ans, r};
      rep(i, 0, n) if(l[i] == -1) ans += dfs(i);
    }
  }
};
```

DFSMatching.h

Description: Simple bipartite matching. r[v] is the left match of right v.
Time: $\mathcal{O}(VE)$

2847a4, 29 lines

```
struct DFSMatching{
  int n, m;
  vector<vi>g;
  vi r, vis;
  DFSMatching(int N, int M): n(N), m(M), g(N){}
  void addEdge(int u, int v){
    g[u].pb(v);
  }
  bool dfs(int u){
    if(vis[u]) return 0;
    vis[u] = 1;
    for(int v : g[u]){
      if(r[v] == -1 or dfs(r[v])){
        r[v] = u;
        return 1;
      }
    }
    return 0;
  }
  pair<int, vi> calc(){
    int ans = 0;
    r.assign(m, -1);
    rep(i, 0, n){
      vis.assign(n, 0);
      ans += dfs(i);
    }
    return {ans, r};
  }
};
```

MinimumVertexCover.h

Description: Minimum vertex cover in a bipartite graph. Returns left vertices as u and right vertices as n+v.
Time: $\mathcal{O}(Esqrt(V))$

"hopcroftKarp.h" 99da56, 23 lines

```
struct MinimumVertexCover : HopcroftKarp{
  using HopcroftKarp::HopcroftKarp;
  vi cover(){
    auto [cnt, mt] = calc();
    vector<bool>left(n, 1), seen(m);
    for(int u : mt) if(u != -1) left[u] = 0;
    vi q, ans;
    rep(i, 0, n) if(left[i]) q.pb(i);
    while(sz(q)){
      int u = q.back();
      q.pop_back();
      left[u] = 1;
      for(int v : g[u]) if(!seen[v] and mt[v] != -1){
        seen[v] = 1;
        q.pb(mt[v]);
      }
    }
    rep(i, 0, n) if(!left[i]) ans.pb(i);
    rep(i, 0, m) if(seen[i]) ans.pb(n + i);
    assert(sz(ans) == cnt);
    return ans;
  }
};
```

WeightedMatching.h

Description: Minimum-cost matching of every row to a distinct column. N<=M.
Time: $\mathcal{O}(N^2M)$

feca40, 21 lines

```
template<class T> pair<T, vi> hungarian(const vector<vector<T>>
    &C){
  int i = sz(C), m = i ? sz(C[0]) : 0, c = 0, s, r;
  vector<T> dist(m), pot(m); vi match(i), rev(m, -1), cols(m),
      pre(m);
  T d = 0, nd, cost = 0;
  while(i--){
    rep(j, 0, m) dist[j] = C[i][j], cols[j] = j, pre[j] = i;
    for(s = 0;;){
      rep(j, s, m){
        c = cols[j], nd = dist[c] - pot[c];
        if(j == s or d > nd) d = nd, swap(cols[s], cols[j]);
      }
      if((r = rev[c = cols[s++]]) == -1) break;
      rep(j, 0, m) if(dist[j] > (nd = C[r][j] - C[r][c] + dist[
          c]))
        dist[j] = nd, pre[j] = r;
    }
    cost += dist[c];
    while(s--) pot[cols[s]] = dist[cols[s]] - d;
    for(; r != i; swap(c, match[r])) r = rev[c] = pre[c];
  }
  return {cost, match};
}
```

GeneralMatching.h

Description: Maximum matching in a general undirected graph.
Time: $\mathcal{O}(N^3)$

ef29b7, 72 lines

```
struct GeneralMatching {
  int n, st;
  vector<vi> g;
  vi mt, even, groupBuf;
  vector<pii> nx;
  GeneralMatching(int n) : n(n), g(n), mt(n), even(n), groupBuf
      (n), nx(n){}
  void addEdge(int a, int b){
    g[a].pb(b); g[b].pb(a);
  }
  int group(int x){
    if(groupBuf[x] == -1 || even[groupBuf[x]] != st) return
        groupBuf[x];
    return groupBuf[x] = group(groupBuf[x]);
  }
  void rematch(int p, int b){
    int d = mt[p]; mt[p] = b;
    if(d == -1 || mt[d] != p) return;
    if(nx[p].S == -1){
      mt[d] = nx[p].F;
      rematch(nx[p].F, d);
    }else{
      rematch(nx[p].F, nx[p].S);
      rematch(nx[p].S, nx[p].F);
    }
  }
  bool augment(){
    even[st] = st; groupBuf[st] = -1; nx[st] = {-1, -1};
    queue<int> q; q.push(st);
    while(!q.empty()){
      int a = q.front(); q.pop();
      for(int b : g[a]){
        if(b == st) continue;
        if(mt[b] == -1){
          mt[b] = a; rematch(a, b);
          return 1;
        }
        if(even[b] == st){
          int x = group(a), y = group(b), z = -1;
          if(x == y) continue;
          while(x != -1 || y != -1){
            if(y != -1) swap(x, y);
            if(nx[x] == pii(a, b)){
              z = x;
              break;
            }
            nx[x] = {a, b};
            x = group(nx[mt[x]].F);
          }
          for(int v : {group(a), group(b)}){
            while(v != z){
              q.push(v); even[v] = st; groupBuf[v] = z;
              v = group(nx[mt[v]].F);
            }
          }
        }else if(even[mt[b]] != st){
          even[mt[b]] = st; nx[b] = {-1, -1};
          nx[mt[b]] = {a, -1}; groupBuf[mt[b]] = b;
          q.push(mt[b]);
        }
      }
    }
    return 0;
  }
  pair<int, vi> calc(){
    fill(all(mt), -1); fill(all(even), -1); fill(all(groupBuf),
        0);
    int ans = 0;
    rep(i, 0, n){
      st = i;
      if(mt[i] == -1) ans += augment();
    }
    return {ans, mt};
```

```
  }
};
```

7.4 DFS algorithms

SCC.h

Description: Iterative Kosaraju SCC. sccid is in reverse topological order.
Time: $\mathcal{O}(V+E)$ ff4399, 44 lines

```
struct SCC{
  int n, cnt = 0;
  vector<vi>g, rg;
  vi sccid;
  SCC(int n): n(n), g(n), rg(n), sccid(n, -1){}
  void add_edge(int u, int v){
    g[u].pb(v);
    rg[v].pb(u);
  }
  int build(){
    vi it(n), order, st;
    vector<bool>vis(n);
    rep(s, 0, n) if(!vis[s]){
      st.pb(s);
      vis[s] = 1;
      while(sz(st)){
        int u = st.back();
        if(it[u] == sz(g[u])){
          order.pb(u);
          st.pop_back();
        }else{
          int v = g[u][it[u]++];
          if(!vis[v]) vis[v] = 1, st.pb(v);
        }
      }
    }
    cnt = 0;
    sccid.assign(n, -1);
    reverse(all(order));
    for(int s : order) if(sccid[s] == -1){
      st.clear();
      st.pb(s);
      sccid[s] = cnt;
      while(sz(st)){
        int u = st.back();
        st.pop_back();
        for(int v : rg[u]) if(sccid[v] == -1) sccid[v] = cnt,
             st.pb(v);
      }
      cnt++;
    }
    for(int &x : sccid) x = cnt - 1 - x;
    return cnt;
  }
};
```

EBCC.h

Description: Bridges and edge-biconnected components of an undirected graph.
Time: $\mathcal{O}(V+E)$ 9859a4, 45 lines

```
struct EBCC{
  int n, m = 0, nbcc = 0, ti = 0;
  vector<vector<pii>>g;
  vi in, low, id, is_bridge;
  EBCC(int n): n(n), g(n){}
  void add_edge(int u, int v){
    g[u].pb({v, m});
    g[v].pb({u, m++});
  }
  int build(){
    nbcc = ti = 0; in.assign(n, -1); low.resize(n); id.assign(n
         , -1);
    is_bridge.assign(m, 0);
    vi it(n), par(n, -1), pe(n, -1), stk;
    rep(s, 0, n) if(in[s] == -1){
      in[s] = low[s] = ti++; stk.pb(s);
      while(sz(stk)){
        int u = stk.back();
        if(it[u] < sz(g[u])){
          auto [v, e] = g[u][it[u]++];
          if(e == pe[u]) continue;
          if(in[v] == -1){
            par[v] = u; pe[v] = e; in[v] = low[v] = ti++; stk.
                 pb(v);
          }else low[u] = min(low[u], in[v]);
        }else{
          stk.pop_back();
          if(pe[u] != -1){
            int p = par[u];
            low[p] = min(low[p], low[u]);
            if(low[u] > in[p]) is_bridge[pe[u]] = 1;
          }
        }
      }
    }
    rep(s, 0, n) if(id[s] == -1){
      id[s] = nbcc++; stk.pb(s);
      while(sz(stk)){
        int u = stk.back(); stk.pop_back();
        for(auto [v, e] : g[u]) if(!is_bridge[e] and id[v] ==
             -1){
          id[v] = id[s]; stk.pb(v);
        }
      }
    }
    return nbcc;
  }
};
```

Dominatortree.h

Description: Immediate dominators from source s. Unreachable nodes get -1.
Time: $\mathcal{O}((V+E)alpha(V))$ 57ee93, 52 lines

```
struct DominatorTree {
  int n, id;
  vector <vector <int>> g, rg, bucket;
  vector <int> sdom, dom, vis, rev, pa, rt, mn, res;
  // dom[s] = s, dom[v] = -1 if s -> v not exists
  int query(int v, int x){
    if(rt[v] == v) return x ? -1 : v;
    int p = query(rt[v], 1);
    if(p == -1) return x ? rt[v] : mn[v];
    if(sdom[mn[v]] > sdom[mn[rt[v]]])
      mn[v] = mn[rt[v]];
    rt[v] = p;
    return x ? p : mn[v];
  }
  void dfs(int v){
    vis[v] = id, rev[id] = v;
    rt[id] = mn[id] = sdom[id] = id, id++;
    for(int u : g[v]){
      if(vis[u] == -1) dfs(u), pa[vis[u]] = vis[v];
      rg[vis[u]].pb(vis[v]);
    }
  }
  void build(int s){
    id = 0;
    for(auto &v : rg) v.clear();
    for(auto &v : bucket) v.clear();
    fill(all(vis), -1); fill(all(dom), -1);
    dfs(s);
    for(int i = id - 1; ~i; --i){
      for(int u : rg[i])
        sdom[i] = min(sdom[i], sdom[query(u, 0)]);
      if(i) bucket[sdom[i]].pb(i);
      for(int u : bucket[i]){
        int p = query(u, 0);
        dom[u] = sdom[p] == i ? i : p;
      }
      if(i) rt[i] = pa[i];
    }
    fill(all(res), -1);
    for(int i = 1; i < id; ++i){
      if(dom[i] != sdom[i]) dom[i] = dom[dom[i]];
    }
    for(int i = 1; i < id; ++i)
      res[rev[i]] = rev[dom[i]];
    res[s] = s;
    for(int i = 0; i < n; ++i) dom[i] = res[i];
  }
  void add_edge(int u, int v){ g[u].pb(v); }
  DominatorTree (int _n) : n(_n), id(0), g(n), rg(n),
    bucket(n), sdom(n), dom(n, -1), vis(n, -1),
    rev(n), pa(n), rt(n), mn(n), res(n){}
};
```

BiconnectedComponents.h

Description: Vertex-biconnected components and the block-cut forest. Original vertices are [0, n), component nodes are [n, n+nbcc).
Time: $\mathcal{O}(V+E)$ f8d87e, 56 lines

```
struct BCC{
  int n, m = 0, nbcc = 0, ti = 0;
  vector<vector<pii>>g;
  vector<vi>tree, comp;
  vi in, low;
  BCC(int n): n(n), g(n){}
  void add_edge(int u, int v){
    g[u].pb({v, m});
    g[v].pb({u, m++});
  }
  void add_comp(int u, int v, vi &stk){
    int id = n + nbcc++;
    tree.pb({}); comp.pb({});
    while(1){
      int x = stk.back(); stk.pop_back();
      tree[id].pb(x), tree[x].pb(id), comp.back().pb(x);
      if(x == v) break;
    }
    tree[id].pb(u), tree[u].pb(id), comp.back().pb(u);
  }
  void build(){
    nbcc = ti = 0; in.assign(n, -1); low.resize(n);
    tree.assign(n, {}); comp.clear(); tree.reserve(n + m);
    vi it(n), par(n, -1), pe(n, -1), stk, dfs;
    rep(s, 0, n) if(in[s] == -1){
      if(g[s].empty()){
        int id = n + nbcc++;
        tree.pb({s}); tree[s].pb(id); comp.pb({s});
        in[s] = low[s] = ti++;
        continue;
      }
      in[s] = low[s] = ti++; stk.pb(s); dfs.pb(s);
      while(sz(dfs)){
        int u = dfs.back();
        if(it[u] < sz(g[u])){
          auto [v, e] = g[u][it[u]++];
```

```
        if(e == pe[u]) continue;
        if(in[v] == -1){
          par[v] = u; pe[v] = e; in[v] = low[v] = ti++;
          stk.pb(v); dfs.pb(v);
        }else low[u] = min(low[u], in[v]);
      }else{
        dfs.pop_back();
        if(pe[u] != -1){
          int p = par[u];
          low[p] = min(low[p], low[u]);
          if(low[u] >= in[p]) add_comp(p, u, stk);
        }
      }
    }
    stk.clear();
  }
 }
 bool is_cut(int u) const { return sz(tree[u]) > 1; }
 const vi &bcc(int id) const { return comp[id]; }
};
```

2sat.h

Description: Calculates a valid assignment to boolean variables a, b, c, ... to a 2-SAT problem, so that an expression of the type $(a||b)\&\&(!a||c)\&\&(d||!b)\&\&...$ becomes true, or reports that it is unsatisfiable. Negated variables are represented by bit-inversions (~x).

Usage:
```
TwoSat ts(number of boolean variables);
ts.either(0, ~3); // Var 0 is true or var 3 is false
ts.setValue(2); // Var 2 is true
ts.atMostOne({0, ~1, 2}); // <= 1 of vars 0, ~1 and 2 are true
ts.solve(); // Returns true iff it is solvable
ts.values[0..N-1] holds the assigned values to the vars
```

Time: $\mathcal{O}(N+E)$, where N is the number of boolean variables, and E is the number of clauses.

b33304, 65 lines

```
struct TwoSat {
  int N;
  vector<vi> gr;
  vi values; // 0 = false, 1 = true

  TwoSat(int n = 0) : N(n), gr(2*n){}

  int addVar(){ // (optional)
    gr.emplace_back();
    gr.emplace_back();
    return N++;
  }

  void either(int f, int j){
    f = max(2*f, -1-2*f);
    j = max(2*j, -1-2*j);
    gr[f].push_back(j^1);
    gr[j].push_back(f^1);
  }
  void setValue(int x){ either(x, x); }

  void atMostOne(const vi& li){ // (optional)
    if(sz(li) <= 1) return;
    int cur = ~li[0];
    rep(i, 2, sz(li)){
      int next = addVar();
      either(cur, ~li[i]);
      either(cur, next);
      either(~li[i], next);
      cur = ~next;
    }
    either(cur, ~li[1]);
  }
  bool solve(){
    vector<vi> rg(2*N); vi vis(2*N), ord, comp(2*N, -1);
    rep(i, 0, 2*N) for(int j : gr[i]) rg[j].pb(i);
    rep(s, 0, 2*N) if(!vis[s]){
      vector<pii> st = {{s, 0}}; vis[s] = 1;
      while(!st.empty()){
        int u = st.back().F;
        if(st.back().S < sz(gr[u])){
          int v = gr[u][st.back().S++];
          if(!vis[v]) vis[v] = 1, st.pb({v, 0});
        }else ord.pb(u), st.pop_back();
      }
    }
    int c = 0;
    for(int oi = sz(ord); oi--;){
      int s = ord[oi]; if(comp[s] != -1) continue;
      vi st = {s}; comp[s] = c;
      while(!st.empty()){
        int u = st.back(); st.pop_back();
        for(int v : rg[u]) if(comp[v] == -1) comp[v] = c, st.pb
            (v);
      }
      ++c;
    }
    values.resize(N);
    rep(i, 0, N){
      if(comp[2*i] == comp[2*i+1]) return 0;
      values[i] = comp[2*i] < comp[2*i+1];
    }
    return 1;
  }
};
```

7.5 Coloring

EdgeColoring.h

Description: Given a simple, undirected graph with max degree D, computes a $(D+1)$-coloring of the edges such that no neighboring edges share a color. (D-coloring is NP-hard, but can be done for bipartite graphs by repeated matchings of max-degree nodes.)

Time: $\mathcal{O}(NM)$

e210e2, 31 lines

```
vi edgeColoring(int N, vector<pii> eds){
  vi cc(N + 1), ret(sz(eds)), fan(N), free(N), loc;
  for(pii e : eds) ++cc[e.first], ++cc[e.second];
  int u, v, ncols = *max_element(all(cc)) + 1;
  vector<vi> adj(N, vi(ncols, -1));
  for(pii e : eds){
    tie(u, v) = e;
    fan[0] = v;
    loc.assign(ncols, 0);
    int at = u, end = u, d, c = free[u], ind = 0, i = 0;
    while(d = free[v], !loc[d] && (v = adj[u][d]) != -1)
      loc[d] = ++ind, cc[ind] = d, fan[ind] = v;
    cc[loc[d]] = c;
    for(int cd = d; at != -1; cd ^= c ^ d, at = adj[at][cd])
      swap(adj[at][cd], adj[end = at][cd ^ c ^ d]);
    while(adj[fan[i]][d] != -1){
      int left = fan[i], right = fan[++i], e = cc[i];
      adj[u][e] = left;
      adj[left][e] = u;
      adj[right][e] = -1;
      free[right] = e;
    }
    adj[u][d] = fan[i];
    adj[fan[i]][d] = u;
    for(int y : {fan[0], u, end})
      for(int& z = free[y] = 0; adj[y][z] != -1; z++);
  }
  rep(i, 0, sz(eds))
    for(tie(u, v) = eds[i]; adj[u][ret[i]] != v;) ++ret[i];
  return ret;
}
```

7.6 Heuristics

MaximalCliques.h

Description: Runs a callback for all maximal cliques in a graph (given as a symmetric bitset matrix; self-edges not allowed). Callback is given a bitset representing the maximal clique.

Time: $\mathcal{O}\left(3^{n/3}\right)$, much faster for sparse graphs

b0d5b1, 12 lines

```
typedef bitset<128> B;
template<class F>
void cliques(vector<B>& eds, F f, B P = ~B(), B X={}, B R={}){
  if(!P.any()){ if(!X.any()) f(R); return; }
  auto q = (P | X)._Find_first();
  auto cands = P & ~eds[q];
  rep(i, 0, sz(eds)) if(cands[i]){
    R[i] = 1;
    cliques(eds, f, P & eds[i], X & eds[i], R);
    R[i] = P[i] = 0; X[i] = 1;
  }
}
```

MaximumClique.h

Description: Fast maximum clique for a symmetric bitset graph, $n <= 200$.

6b21e8, 43 lines

```
typedef vector<bitset<200>> vb;
struct Maxclique {
  double limit=.025, pk=0;
  struct Vertex { int i, d=0; };
  using vv=vector<Vertex>;
  vb e; vv V; vector<vi> C; vi qmax, q, steps, old;
  void init(vv &r){
    if(r.empty()) return;
    for(auto &v:r) v.d=0;
    for(auto &v:r) for(auto u:r) v.d+=e[v.i][u.i];
    sort(all(r), [](auto a, auto b){return a.d>b.d;});
    int mx=r[0].d;
    rep(i, 0, sz(r)) r[i].d=min(i, mx)+1;
  }
  void expand(vv &R, int lev=1){
    steps[lev]+=steps[lev-1]-old[lev]; old[lev]=steps[lev-1];
    while(sz(R)){
      if(sz(q)+R.back().d<=sz(qmax)) return;
      q.pb(R.back().i); vv T;
      for(auto v:R) if(e[R.back().i][v.i]) T.pb({v.i});
      if(sz(T)){
        if((double)steps[lev]++/++pk<limit) init(T);
        int j=0, mx=1, mn=max<int>(sz(qmax)-sz(q)+1, 1);
        C[1].clear(), C[2].clear();
        for(auto v:T){
          int k=1;
          while(any_of(all(C[k]), [&](int i){return e[v.i][i];}
              )) ++k;
          if(k>mx) mx=k, C[mx+1].clear();
          if(k<mn) T[j++].i=v.i;
          C[k].pb(v.i);
        }
        if(j) T[j-1].d=0;
        rep(k, mn, mx+1) for(int i:C[k]) T[j].i=i, T[j++].d=k;
        expand(T, lev+1);
      } else if(sz(q)>sz(qmax)) qmax=q;
      q.pop_back(), R.pop_back();
    }
  }
  vi maxClique(){ if(V.empty()) return {}; init(V), expand(V);
      return qmax; }
```

```
  Maxclique(vb e):e(e), C(sz(e)+2), steps(C.size()), old(steps)
      {
    rep(i, 0, sz(e)) V.pb({i});
  }
};
```

MaximumIndependentSet.h

Description: To obtain a maximum independent set of a graph, find a max clique of the complement. If the graph is bipartite, see MinimumVertexCover.

7.7 Trees

CompressTree.h

Description: Given a rooted tree and a subset S of nodes, compute the minimal subtree that contains all the nodes by adding all (at most $|S|-1$) pairwise LCA's and compressing edges. Returns a list of (par, orig_index) representing a tree rooted at 0. The root points to itself.
Time: $\mathcal{O}(|S|\log|S|)$

"LCA.h" bb56cc, 22 lines

```
typedef vector<pair<int, int>> vpi;
vpi compressTree(LCA& lca, const vi& subset){
  if(subset.empty()) return {};
  static vi rev; rev.resize(sz(lca.time));
  vi li = subset, &T = lca.time;
  auto cmp = [&](int a, int b){ return T[a] < T[b]; };
  sort(all(li), cmp);
  int m = sz(li)-1;
  rep(i, 0, m){
    int a = li[i], b = li[i+1];
    li.push_back(lca.lca(a, b));
  }
  sort(all(li), cmp);
  li.erase(unique(all(li)), li.end());
  rep(i, 0, sz(li)) rev[li[i]] = i;
  vpi ret = {pii(0, li[0])};
  rep(i, 0, sz(li)-1){
    int a = li[i], b = li[i+1];
    ret.emplace_back(rev[lca.lca(a, b)], b);
  }
  return ret;
}
```

HLD.h

Description: Decomposes a tree into vertex disjoint heavy paths and light edges such that the path from any leaf to the root contains at most log(n) light edges. Code does additive modifications and max queries, but can support commutative segtree modifications/queries on paths and subtrees. Takes as input the full adjacency list. VALS_EDGES being true means that values are stored in the edges, as opposed to the nodes. All values initialized to the segtree default. Root must be 0.
Time: $\mathcal{O}\left((\log N)^2\right)$

"../data-structures/LazySegmentTree.h" 73ed04, 46 lines

```
template <bool VALS_EDGES> struct HLD {
  int N, tim = 0;
  vector<vi> adj;
  vi par, siz, rt, pos;
  Node *tree;
  HLD(vector<vi> adj_)
    : N(sz(adj_)), adj(adj_), par(N, -1), siz(N, 1),
      rt(N), pos(N), tree(new Node(0, N)){ dfsSz(0); dfsHld(0);
          }
  void dfsSz(int v){
    for(int& u : adj[v]){
      adj[u].erase(find(all(adj[u]), v));
      par[u] = v;
      dfsSz(u);
      siz[v] += siz[u];
      if(siz[u] > siz[adj[v][0]]) swap(u, adj[v][0]);
    }
  }
  void dfsHld(int v){
    pos[v] = tim++;
    for(int u : adj[v]){
      rt[u] = (u == adj[v][0] ? rt[v] : u);
      dfsHld(u);
    }
  }
  template <class B> void process(int u, int v, B op){
    for(;; v = par[rt[v]]){
      if(pos[u] > pos[v]) swap(u, v);
      if(rt[u] == rt[v]) break;
      op(pos[rt[v]], pos[v] + 1);
    }
    op(pos[u] + VALS_EDGES, pos[v] + 1);
  }
  void modifyPath(int u, int v, ll val){
    process(u, v, [&](int l, int r){ tree->add(l, r, val); });
  }
  ll queryPath(int u, int v){ // Modify depending on problem
    ll res = -inf;
    process(u, v, [&](int l, int r){
        res = max(res, tree->query(l, r));
    });
    return res;
  }
  ll querySubtree(int v){ // modifySubtree is similar
    return tree->query(pos[v] + VALS_EDGES, pos[v] + siz[v]);
  }
};
```

LinkCutTree.h

Description: 1-based link-cut tree with path and rooted-subtree aggregates. Val and SVal need + and -; Val also needs reverse-order aggregation.
Usage: access(u): expose root-u path, splay u, and clear its right child.
chroot(u): make u the represented-tree root.
split(u, v): expose path u-v in v; its aggregate is o[v].sum.
vir stores virtual-child aggregates; access moves preferred children
into/out of vir. sub includes both aux children, vir, and sv.
Time: $\mathcal{O}(logN)$ amortized

8251ab, 95 lines

```
// 1-based
// == PART HASH ==
template <typename Val, typename SVal> struct LCT {
  struct node {
    int pa, ch[2]; bool rev; int size;
    Val v, sum, rsum; SVal sv, sub, vir;
    node() : pa{0}, ch{0, 0}, rev{false}, size{1}, v{},
      sum{}, rsum{}, sv{}, sub{}, vir{} {}
  };
#define cur o[u]
#define lc cur.ch[0]
#define rc cur.ch[1]
  vector<node> o;
  bool is_root(int u) const {
    return o[cur.pa].ch[0]!=u && o[cur.pa].ch[1]!=u; }
  bool is_rch(int u) const {
    return o[cur.pa].ch[1] == u && !is_root(u); }
  void down(int u){
    for(int c : {lc, rc}) if(c){
      if(cur.rev) set_rev(c);
    }
    cur.rev = false;
  }
  void up(int u){
    cur.sum = o[lc].sum + cur.v + o[rc].sum;
    cur.rsum = o[rc].rsum + cur.v + o[lc].rsum;
    cur.sub = cur.vir + o[lc].sub + o[rc].sub + cur.sv;
    cur.size = o[lc].size + o[rc].size + 1;
  }
  void set_rev(int u){
    swap(lc, rc), swap(cur.sum, cur.rsum);
    cur.rev ^= 1;
  }
// == PART HASH ==
  void rotate(int u){
    int f = cur.pa, g = o[f].pa, l = is_rch(u);
    if(cur.ch[l ^ 1]) o[cur.ch[l ^ 1]].pa = f;
    if(not is_root(f)) o[g].ch[is_rch(f)] = u;
    o[f].ch[l] = cur.ch[l ^ 1], cur.ch[l ^ 1] = f;
    cur.pa = g, o[f].pa = u; up(f);
  }
  vector<int> stk;
  void splay(int u){
    stk.clear(); stk.pb(u);
    while(not is_root(stk.back()))
      stk.push_back(o[stk.back()].pa);
    while(not stk.empty())
      down(stk.back()), stk.pop_back();
    for(int f = cur.pa; not is_root(u); f = cur.pa){
      if(!is_root(f))
        rotate(is_rch(u) == is_rch(f) ? f : u);
      rotate(u);
    }
    up(u);
  }
  void access(int x){
    for(int u = x, last = 0; u; u = cur.pa){
      splay(u);
      cur.vir = cur.vir + o[rc].sub - o[last].sub;
      rc = last; up(last = u);
    }
    splay(x);
  }
  int find_root(int u){
    int la = 0;
    for(access(u); u; u = lc) down(la = u);
    splay(la);
    return la;
  }
  void split(int x, int y){ chroot(x); access(y); }
  void chroot(int u){ access(u); set_rev(u); }
// == PART HASH ==
  LCT(int n = 0) : o(n + 1){ o[0].size = 0; }
  void set_val(int u, const Val &v){
    access(u); cur.v = v; up(u); }
  void set_sval(int u, const SVal &v){
    access(u); cur.sv = v; up(u); }
  Val query(int x, int y){
    split(x, y); return o[y].sum; }
  SVal subtree(int p, int u){
    chroot(p); access(u); return cur.vir + cur.sv; }
  bool connected(int u, int v){
    return find_root(u) == find_root(v); }
  void link(int x, int y){
    chroot(x); assert(find_root(y) != x); access(y);
    o[y].vir = o[y].vir + o[x].sub;
    up(o[x].pa = y);
  }
  void cut(int x, int y){
    split(x, y); assert(o[y].ch[0] == x && !o[x].ch[1]);
    o[y].ch[0] = o[x].pa = 0; up(y); }
#undef cur
```

```
#undef lc
#undef rc
};
```

DirectedMST.h

Description: Finds a minimum spanning tree/arborescence of a directed graph, given a root node. If no MST exists, returns -1.
Time: $\mathcal{O}(E \log V)$

"../data-structures/UnionFindRollback.h" 39e620, 60 lines

```
struct Edge { int a, b; ll w; };
struct Node {
  Edge key;
  Node *l, *r;
  ll delta;
  void prop(){
    key.w += delta;
    if(l) l->delta += delta;
    if(r) r->delta += delta;
    delta = 0;
  }
  Edge top(){ prop(); return key; }
};
Node *merge(Node *a, Node *b){
  if(!a || !b) return a ?: b;
  a->prop(), b->prop();
  if(a->key.w > b->key.w) swap(a, b);
  swap(a->l, (a->r = merge(b, a->r)));
  return a;
}
void pop(Node*& a){ a->prop(); a = merge(a->l, a->r); }

pair<ll, vi> dmst(int n, int r, vector<Edge>& g){
  RollbackUF uf(n);
  vector<Node*> heap(n);
  for(Edge e : g) heap[e.b] = merge(heap[e.b], new Node{e});
  ll res = 0;
  vi seen(n, -1), path(n), par(n);
  seen[r] = r;
  vector<Edge> Q(n), in(n, {-1, -1}), comp;
  deque<tuple<int, int, vector<Edge>>> cycs;
  rep(s, 0, n){
    int u = s, qi = 0, w;
    while(seen[u] < 0){
      if(!heap[u]) return {-1, {}};
      Edge e = heap[u]->top();
      heap[u]->delta -= e.w, pop(heap[u]);
      Q[qi] = e, path[qi++] = u, seen[u] = s;
      res += e.w, u = uf.find(e.a);
      if(seen[u] == s){
        Node* cyc = 0;
        int end = qi, time = uf.time();
        do cyc = merge(cyc, heap[w = path[--qi]]);
        while(uf.join(u, w));
        u = uf.find(u), heap[u] = cyc, seen[u] = -1;
        cycs.push_front({u, time, {&Q[qi], &Q[end]}});
      }
    }
    rep(i, 0, qi) in[uf.find(Q[i].b)] = Q[i];
  }

  for(auto& [u, t, comp] : cycs){ // restore sol (optional)
    uf.rollback(t);
    Edge inEdge = in[u];
    for(auto& e : comp) in[uf.find(e.b)] = e;
    in[uf.find(inEdge.b)] = inEdge;
  }
  rep(i, 0, n) par[i] = in[i].a;
  return {res, par};
}
```

7.8 Math

7.8.1 Number of Spanning Trees

Create an $N \times N$ matrix `mat`, and for each edge $a \to b \in G$, do `mat[a][b]--, mat[b][b]++` (and `mat[b][a]--, mat[a][a]++` if G is undirected). Remove the ith row and column and take the determinant; this yields the number of directed spanning trees rooted at i (if G is undirected, remove any row/column).

7.8.2 Erdős–Gallai theorem

A simple graph with node degrees $d_1 \geq \cdots \geq d_n$ exists iff $d_1 + \cdots + d_n$ is even and for every $k = 1 \ldots n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(d_i, k).$$

Geometry (8)

8.1 Geometry

GeometryTemplate.cpp

Description: Common 2D geometry primitives for pair-based points, including vector arithmetic, orientation, segment tests, line intersection, projection, and reflection.
Time: $\mathcal{O}(1)$ per operation

407dd0, 60 lines

```
using ld = double;
using pdd = pair<ld, ld>;
#define X first
#define Y second
const ld eps = 1e-9;

#define TEMP template<class T>
#define ptt pair<T, T>
TEMP ptt operator+(ptt a, ptt b)
{ return {a.X + b.X, a.Y + b.Y}; }
TEMP ptt operator-(ptt a, ptt b)
{ return {a.X - b.X, a.Y - b.Y}; }
TEMP ptt operator*(T i, ptt v)
{ return {i * v.X, i * v.Y}; }
TEMP ptt operator*(ptt v, T i)
{ return {i * v.X, i * v.Y}; }
TEMP pdd operator/(ptt v, ld i)
{ return {v.X / i, v.Y / i}; }
TEMP T dot(ptt a, ptt b)
{ return a.X * b.X + a.Y * b.Y; }
TEMP T cross(ptt a, ptt b)
{ return a.X * b.Y - a.Y * b.X; }
TEMP T abs2(ptt v)
{ return v.X * v.X + v.Y * v.Y; };
TEMP ld abs(ptt v)
{ return sqrt(abs2(v)); };
int sgn(ll v)
{ return v > 0 ? 1 : (v < 0 ? -1 : 0); }
int sgn(ld v){ return v > eps ? 1 : ( v < -eps ? -1 : 0); }
TEMP int ori(ptt a, ptt b, ptt c)
{ return sgn(cross(b - a, c - a)); }
TEMP bool collinearity(ptt a, ptt b, ptt c)
{ return ori(a, b, c) == 0; }
TEMP bool btw(ptt p, ptt a, ptt b)
{ return collinearity(p, a, b) && sgn(dot(a - p, b - p)) <= 0;
    }
TEMP bool seg_strict_intersect(ptt p1, ptt p2, ptt p3, ptt p4){
  return ori(p1, p2, p3) * ori(p1, p2, p4) < 0 && ori(p3, p4,
      p1) * ori(p3, p4, p2) < 0;
}
TEMP bool seg_intersect(ptt p1, ptt p2, ptt p3, ptt p4){
  if(btw(p1, p3, p4) || btw(p2, p3, p4) || btw(p3, p1, p2) ||
      btw(p4, p1, p2))
    return true;
  return ori(p1, p2, p3) * ori(p1, p2, p4) < 0 &&
    ori(p3, p4, p1) * ori(p3, p4, p2) < 0;
}
pdd intersect(pdd p1, pdd p2, pdd p3, pdd p4){
  ld a123 = cross(p2 - p1, p3 - p1);
  ld a124 = cross(p2 - p1, p4 - p1);
  return (p4 * a123 - p3 * a124) / (a123 - a124);
}
TEMP ptt perp(ptt p1)
{ return {-p1.Y, p1.X}; }
pdd projection(pdd p1, pdd p2, pdd p3)
{ return p1 + (p2 - p1) * dot(p3 - p1, p2 - p1) / abs2(p2 - p1)
    ; }
pdd reflection(pdd p1, pdd p2, pdd p3)
{ return p3 + perp(p2 - p1) * cross(p3 - p1, p2 - p1) / abs2(p2
     - p1) * 2.0; }
pdd linearTransformation(pdd p0, pdd p1, pdd q0, pdd q1, pdd r)
    {
  pdd dp = p1 - p0, dq = q1 - q0, num(cross(dp, dq), dot(dp, dq
      ));
  return q0 + pdd(cross(r - p0, num), dot(r - p0, num)) / abs2(
      dp);
} // from line p0—p1 to q0—q1, apply to r
```

CycTsearch.cpp

Description: Given a predicate over adjacent values of a cyclically shifted unimodal sequence, returns an index for which pred(x, index) is false for every x.
Time: $\mathcal{O}(\log N)$

9017cc, 10 lines

```
int cyc_tsearch(int n, auto pred) {
  if (n == 1) return 0;
  int l = 0, r = n; bool rv = pred(1, 0);
  while (r - l > 1) {
    int m = (l + r) / 2;
    if (pred(0, m) ? rv: pred(m, (m + 1) % n)) r = m;
    else l = m;
  }
  return pred(l, r % n) ? l : r % n;
}
```

ConvexHull.cpp

Description: Computes the indices of the strict convex hull of a set of 2D points. Collinear points on hull edges are discarded. Use pll for exact integer hulls.
Time: $\mathcal{O}(N \log N)$

32febc, 20 lines

```
template<class P> vector<int> getConvexHull(vector<P>& pts){
  if(sz(pts) <= 1) return vector<int>(sz(pts), 0);
  vector<int> id(sz(pts));
  iota(all(id), 0);
  sort(all(id), [&](int x, int y){ return pts[x] < pts[y]; });
  vector<int> hull;
  for(int tt = 0; tt < 2; tt++){
    int sz = sz(hull);
    for(int j : id){
      P p = pts[j];
      while(sz(hull) - sz >= 2 && ori(pts[hull.end()[-2]], pts[
          hull.back()], p) <= 0)
        hull.pop_back();
```

```
      hull.pb(j);
    }
    hull.pop_back();
    reverse(all(id));
  }
  if(sz(hull) == 2 && pts[hull[0]] == pts[hull[1]]) hull.
      pop_back();
  return hull;
}
```

HullDiameter.cpp

Description: Squared diameter (farthest pair) of a convex polygon given in CCW order without collinear points.
Time: $\mathcal{O}(n)$

dc285a, 9 lines

```
pair<ll, pii> hullDiameter(const vector<pll>& h){
  int n = sz(h), j = n < 2 ? 0 : 1;
  pair<ll, pii> res{0, {0, 0}};
  rep(i, 0, j) for(;; j = (j + 1) % n){
    res = max(res, {abs2(h[i] - h[j]), {i, j}});
    if(cross(h[(j + 1) % n] - h[j], h[i + 1] - h[i]) >= 0)
        break;
  }
  return res;
}
```

PolarAngleComp.cpp

Description: Compares integer vectors by polar angle without trigonometry. Equal directions can be ordered by squared length or reported as equivalent.
Time: $\mathcal{O}(1)$

808e89, 10 lines

```
// -1: a // b (if same), 0/1: a < b
int cmp(pll a, pll b, bool same = true){
#define is_neg(k) (sgn(k.Y) < 0 || (sgn(k.Y) == 0 && sgn(k.X) <
      0))
  int A = is_neg(a), B = is_neg(b);
  if(A != B)
    return A < B;
  if(sgn(cross(a, b)) == 0)
    return same ? abs2(a) < abs2(b) : -1;
  return sgn(cross(a, b)) > 0;
}
```

MinkowskiSum.cpp

Description: Computes the Minkowski sum of two convex polygons by merging their edge directions after rotating both polygons to a canonical starting vertex.
Time: $\mathcal{O}(N+M)$

dfb169, 18 lines

```
TEMP void reorder_poly(vector<ptt>& pts){
  rotate(pts.begin(), min_element(all(pts),
    [&](ptt x, ptt y){
    return x.Y != y.Y ? x.Y < y.Y : x.X < y.X; }),
    pts.end());
}
TEMP vector<ptt> minkowski(vector<ptt> P, vector<ptt> Q){
  reorder_poly(P); reorder_poly(Q);
  int psz = P.size(), qsz = Q.size();
  P.pb(P[0]); P.pb(P[1]); Q.pb(Q[0]); Q.pb(Q[1]);
  vector<ptt> ans; int i = 0, j = 0;
  while (i < psz || j < qsz) {
    ans.pb(P[i] + Q[j]);
    int t = sgn(cross(P[i + 1]-P[i], Q[j + 1]-Q[j]));
    if(t >= 0) i++; if(t <= 0) j++;
  }
  return ans;
}
```

CircleConvexIntersection.cpp

Description: Computes the area of intersection between a circle and a polygon whose vertices are given in boundary order.
Time: $\mathcal{O}(N)$

8df18f, 26 lines

```
const double PI = acos(-1);
double _area(pdd pa, pdd pb, double r){
  if(abs(pa)<abs(pb)) swap(pa, pb);
  if(abs(pb)<eps) return 0;
  double S, h, theta;
  double a=abs(pb),b=abs(pa),c=abs(pb-pa);
  double cosB = dot(pb,pb-pa) / a / c, B = acos(clamp(cosB,
      -1.0, 1.0));
  double cosC = dot(pa,pb) / a / b, C = acos(clamp(cosC, -1.0,
      1.0));
  if(a > r){
    S = (C/2)*r*r;
    h = a*b*sin(C)/c;
    if (h < r && B < PI/2) S -= (acos(h/r)*r*r - h*sqrt(r*r-h*h
        ));
  }
  else if(b > r){
    theta = PI - B - asin(sin(B)/r*a);
    S = .5*a*r*sin(theta) + (C-theta)/2*r*r;
  }
  else S = .5*sin(C)*a*b;
  return S;
}
double areaPolyCircle(const vector<pdd> poly,const pdd &O,const
     double r){
  double S=0;
  for(int i=0;i<sz(poly);++i)
    S+=_area(poly[i]-O,poly[(i+1)%sz(poly)]-O,r)*ori(O,poly[i],
        poly[(i+1)%sz(poly)]);
  return fabs(S);
}
```

CircleCircleIntersection.cpp

Description: Computes the intersection points of two non-concentric circles. Returns false when the circles are disjoint or one lies strictly inside the other.
Time: $\mathcal{O}(1)$

f7a2fe, 10 lines

```
bool CCinter(Cir &a, Cir &b, pdd &p1, pdd &p2) {
  pdd o1 = a.O, o2 = b.O;
  double r1 = a.R, r2 = b.R, d2 = abs2(o1 - o2), d = sqrt(d2);
  if(d < max(r1, r2) - min(r1, r2) || d > r1 + r2) return 0;
  pdd u = (o1 + o2) * 0.5 + (o1 - o2) * ((r2 * r2 - r1 * r1) /
      (2 * d2));
  double A = sqrt((r1 + r2 + d) * (r1 - r2 + d) * (r1 + r2 - d)
      * (-r1 + r2 + d));
  pdd v = pdd(o1.Y - o2.Y, -o1.X + o2.X) * A / (2 * d2);
  p1 = u + v, p2 = u - v;
  return 1;
}
```

CircleCircleTangent.cpp

Description: Returns the common tangent lines of two distinct circles. Use sign1 = 1 for outer tangents and sign1 = -1 for inner tangents.
Time: $\mathcal{O}(1)$

c51d90, 21 lines

```
vector<Line> CCtang( const Cir& c1 , const Cir& c2 , int sign1
    ){
  vector<Line> ret;
  double d_sq = abs2( c1.O - c2.O );
  if (sgn(d_sq) == 0) return ret;
  double d = sqrt(d_sq);
  pdd v = (c2.O - c1.O) / d;
  double c = (c1.R - sign1 * c2.R) / d; // cos t
  if (c * c > 1) return ret;
  double h = sqrt(max( 0.0, 1.0 - c * c)); // sin t
  for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
    pdd n = pdd(v.X * c - sign2 * h * v.Y,
        v.Y * c + sign2 * h * v.X);
    pdd p1 = c1.O + n * c1.R;
    pdd p2 = c2.O + n * (c2.R * sign1);
    if (sgn(p1.X - p2.X) == 0 and
        sgn(p1.Y - p2.Y) == 0)
      p2 = p1 + perp(c2.O - c1.O);
    ret.pb(Line(p1, p2));
  }
  return ret;
}
```

ConvexLineIntersection.cpp

Description: Finds the sides or vertex where an infinite directed line intersects a convex polygon. Returns -1, -1 when there is no intersection.
Time: $\mathcal{O}(\log N)$

7d8485, 32 lines

```
int TangentDir(vector<pll> &C, pll dir) {
  return cyc_tsearch(sz(C), [&](int a, int b) {
    return cross(dir, C[a]) > cross(dir, C[b]);
  });
}
#define cmpL(i) sgn(cross(C[i] - a, b - a))
pii lineHull(pll a, pll b, vector<pll> &C) {
  int A = TangentDir(C, a - b);
  int B = TangentDir(C, b - a);
  int n = sz(C);
  if (cmpL(A) < 0 || cmpL(B) > 0)
    return pii(-1, -1); // no collision
  auto gao = [&](int l, int r) {
    for (int t = l; (l + 1) % n != r; ) {
      int m = ((l + r + (l < r ? 0 : n)) / 2) % n;
      (cmpL(m) == cmpL(t) ? l : r) = m;
    }
    return (l + !cmpL(r)) % n;
  };
  pii res = pii(gao(B, A), gao(A, B)); // (i, j)
  if (res.X == res.Y) // touching the corner i
    return pii(res.X, -1);
  if (!cmpL(res.X) && !cmpL(res.Y)) // along side i, i+1
    switch ((res.X - res.Y + n + 1) % n) {
      case 0: return pii(res.X, res.X);
      case 2: return pii(res.Y, res.Y);
    }
  /* crossing sides (i, i+1) and (j, j+1)
   crossing corner i is treated as side (i, i+1)
   returned in the same order as the line hits the convex */
  return res;
} // convex cut: (r, l]
```

CircleLineIntersection.cpp

Description: Returns the zero, one, or two intersection points between a circle and the infinite line through points a and b.
Time: $\mathcal{O}(1)$

9183db, 8 lines

```
vector<pdd> circleLineIntersection(pdd c, double r, pdd a, pdd
    b) {
  pdd p = a + (b - a) * dot(c - a, b - a) / abs2(b - a);
  double s = cross(b - a, c - a), h2 = r * r - s * s / abs2(b -
      a);
  if (sgn(h2) < 0) return {};
  if (sgn(h2) == 0) return {p};
  pdd h = (b - a) / abs(b - a) * sqrt(h2);
  return {p - h, p + h};
}
```

PointInCircle.cpp

Description: Uses an exact determinant to test whether an integer point lies strictly inside the circumcircle of a counter-clockwise integer triangle.
Time: $\mathcal{O}(1)$

ecf954, 7 lines

```
// return q's relation with circumcircle of tri(p[0],p[1],p[2])
bool in_cc(const array<pll, 3> &p, pll q) {
  __int128 det = 0;
  for (int i = 0; i < 3; ++i)
    det += __int128(abs2(p[i]) - abs2(q)) * cross(p[(i + 1) %
        3] - q, p[(i + 2) % 3] - q);
  return det > 0; // in: >0, on: =0, out: <0
}
```

PointInConvex.cpp

Description: Tests whether a point lies inside a convex polygon in logarithmic time. The strict flag controls whether boundary points are accepted.
Time: $\mathcal{O}(\log N)$

fe4934, 13 lines

```
bool PointInConvex(const vector<pll> &C, pll p, bool strict =
    true) {
  int a = 1, b = sz(C) - 1, r = !strict;
  if (sz(C) == 0) return false;
  if (sz(C) < 3) return r && btw(p, C[0], C.back());
  if (ori(C[0], C[a], C[b]) > 0) swap(a, b);
  if (ori(C[0], C[a], p) >= r || ori(C[0], C[b], p) <= -r)
    return false;
  while (abs(a - b) > 1) {
    int c = (a + b) / 2;
    (ori(C[0], C[c], p) > 0 ? b : a) = c;
  }
  return ori(C[a], C[b], p) < r;
}
```

HalfPlaneIntersection.cpp

Description: Intersects directed half-planes and returns the boundary lines of the resulting convex region. Assumes the intersection is non-empty and checks strict inclusion.
Time: $\mathcal{O}(N \log N)$

3d7062, 32 lines

```
pll area_pair(Line a, Line b)
{ return pll(cross(a.Y - a.X, b.X - a.X), cross(a.Y - a.X, b.Y
    - a.X)); }
bool isin(Line l0, Line l1, Line l2) {
  // Check inter(l1, l2) strictly in l0
  auto [a02X, a02Y] = area_pair(l0, l2);
  auto [a12X, a12Y] = area_pair(l1, l2);
  if (a12X - a12Y < 0) a12X *= -1, a12Y *= -1;
  return (__int128) a02Y * a12X - (__int128) a02X * a12Y > 0;
}
/* Having solution, check size > 2 */
/* —^— Line.X —^— Line.Y —^— */
vector<Line> halfPlaneInter(vector<Line> arr) {
  sort(all(arr), [&](Line a, Line b) -> int {
    if (cmp(a.Y - a.X, b.Y - b.X, 0) != -1)
      return cmp(a.Y - a.X, b.Y - b.X, 0);
    return ori(a.X, a.Y, b.Y) < 0;
  });
  deque<Line> dq(1, arr[0]);
  auto pop_back = [&](int t, Line p) {
    while (sz(dq) >= t && !isin(p, dq[sz(dq) - 2], dq.back()))
      dq.pop_back();
  };
  auto pop_front = [&](int t, Line p) {
    while (sz(dq) >= t && !isin(p, dq[0], dq[1]))
      dq.pop_front();
  };
  for (auto p : arr)
    if (cmp(dq.back().Y - dq.back().X, p.Y - p.X, 0) != -1)
      pop_back(2, p), pop_front(2, p), dq.pb(p);
  pop_back(3, dq[0]), pop_front(3, dq.back());
  return vector<Line>(all(dq));
}
```

HPIGeneralLine.cpp

Description: Exact integer-line helpers for half-plane intersection. Lines are stored as $ax + by + c \le 0$, while intersections are returned as floating-point points.
Time: $\mathcal{O}(1)$ per operation

aa280c, 25 lines

```
// replace corresponding functions in vector HPI
using i128 = __int128;
int sgn(i128 v){ return (v > 0) - (v < 0); }
struct LN {
  ll a, b, c; // ax + by + c <= 0
  pll dir() const { return pll(a, b); }
  LN(ll ta, ll tb, ll tc) : a(ta), b(tb), c(tc) {}
  LN(pll S, pll T): a((T-S).Y), b(-(T-S).X), c(cross(T,S)) {}
};
pdd intersect(LN A, LN B) {
  double c = cross(A.dir(), B.dir());
  i128 a = i128(A.c) * B.a - i128(B.c) * A.a;
  i128 b = i128(A.c) * B.b - i128(B.c) * A.b;
  return pdd(-b / c, a / c);
}
bool isin(LN l, LN A, LN B) {
  i128 c = cross(A.dir(), B.dir());
  i128 a = i128(A.c) * B.a - i128(B.c) * A.a;
  i128 b = i128(A.c) * B.b - i128(B.c) * A.b;
  return sgn(a * l.b - b * l.a + c * l.c) * sgn(c) < 0;
}
int cmp(LN a, LN b) {
  if (int c = cmp(a.dir(), b.dir(), false); c != -1) return c;
  return i128(abs(b.a) + abs(b.b)) * a.c > i128(abs(a.a) + abs(
      a.b)) * b.c;
}
```

MinimumEnclosingCircle.cpp

Description: Randomized incremental algorithm for the minimum enclosing circle of a non-empty point set. Requires a circumcenter routine for three boundary points.
Time: $\mathcal{O}(N)$ expected

8be940, 23 lines

```
pair<pdd, ld> circumcenter(pdd A, pdd B, pdd C){
  pdd b = C - A, c = B - A, o = A + perp(b * abs2(c) - c * abs2
      (b)) / cross(b, c) / 2;
  return {o, abs(o - A)};
}
pair<pdd, ld> MinimumEnclosingCircle(vector<pdd> &pts){
  shuffle(all(pts), mt19937(1));
  pdd c = pts[0];
  ld r = 0;
  for(int i = 1; i < sz(pts); i++){
    if(abs(pts[i] - c) <= r) continue;
    c = pts[i]; r = 0;
    for(int j = 0; j < i; j++){
      if(abs(pts[j] - c) <= r) continue;
      c = (pts[i] + pts[j]) / 2;
      r = abs(pts[i] - c);
      for(int k = 0; k < j; k++){
        if(abs(pts[k] - c) > r)
          tie(c, r) = circumcenter(pts[i], pts[j], pts[k]);
      }
    }
  }
  return {c, r};
}
```

3DPoint.cpp

Description: Basic 3D point and vector operations, including dot and cross products, signed volume, spherical angles, projection onto a plane, and rotation around an axis.
Time: $\mathcal{O}(1)$ per operation

7d4936, 47 lines

```
struct Point {
  double x, y, z;
  Point(double _x = 0, double _y = 0, double _z = 0): x(_x), y(
      _y), z(_z){}
  Point(pdd p) { x = p.X, y = p.Y, z = abs2(p); }
};
Point operator-(Point p1, Point p2)
{ return Point(p1.x - p2.x, p1.y - p2.y, p1.z - p2.z); }
Point operator+(Point p1, Point p2)
{ return Point(p1.x + p2.x, p1.y + p2.y, p1.z + p2.z); }
Point operator*(Point p1, double v)
{ return Point(p1.x * v, p1.y * v, p1.z * v); }
Point operator/(Point p1, double v)
{ return Point(p1.x / v, p1.y / v, p1.z / v); }
Point cross(Point p1, Point p2)
{ return Point(p1.y * p2.z - p1.z * p2.y, p1.z * p2.x - p1.x *
    p2.z, p1.x * p2.y - p1.y * p2.x); }
double dot(Point p1, Point p2)
{ return p1.x * p2.x + p1.y * p2.y + p1.z * p2.z; }
double abs2(Point a){ return dot(a, a); }
double abs(Point a)
{ return sqrt(dot(a, a)); }
Point cross3(Point a, Point b, Point c)
{ return cross(b - a, c - a); }
double area(Point a, Point b, Point c)
{ return abs(cross3(a, b, c)); }
double volume(Point a, Point b, Point c, Point d)
{ return dot(cross3(a, b, c), d - a); }
//Azimuthal angle (longitude) to x-axis in interval [-pi, pi]
double phi(Point p) { return atan2(p.y, p.x); }
//Zenith angle (latitude) to the z-axis in interval [0, pi]
double theta(Point p) { return atan2(sqrt(p.x * p.x + p.y * p.y
    ), p.z); }
Point masscenter(Point a, Point b, Point c, Point d)
{ return (a + b + c + d) / 4; }
pdd proj(Point a, Point b, Point c, Point u) {
// proj. u to the plane of a, b, and c
  Point e1 = b - a;
  Point e2 = c - a;
  e1 = e1 / abs(e1);
  e2 = e2 - e1 * dot(e2, e1);
  e2 = e2 / abs(e2);
  Point p = u - a;
  return pdd(dot(p, e1), dot(p, e2));
}
Point rotate_around(Point p, double angle, Point axis) {
  double s = sin(angle), c = cos(angle);
  Point u = axis / abs(axis);
  return u * dot(u, p) * (1 - c) + p * c + cross(u, p) * s;
}
```

ConvexHull3D.cpp

Description: Incrementally constructs the triangular faces of a 3D convex hull and provides its coplanar face count, volume, and point-to-face distance.
Time: $\mathcal{O}(N^2)$ construction

651397, 65 lines

```
struct convex_hull_3D {
struct Face {
  int a, b, c;
  Face(int ta, int tb, int tc): a(ta), b(tb), c(tc) {}
}; // return the faces with pt indexes
vector<Face> res;
vector<Point> P;
convex_hull_3D(const vector<Point> &_P): res(), P(_P) {
```

```
// all points coplanar case will WA, O(n^2)
  int n = sz(P);
  if (n <= 2) return; // be careful about edge case
  // ensure first 4 points are not coplanar
  swap(P[1], *find_if(all(P), [&](auto p) { return sgn(abs2(P
      [0] - p)) != 0; }));
  swap(P[2], *find_if(all(P), [&](auto p) { return sgn(abs2(
      cross3(p, P[0], P[1]))) != 0; }));
  swap(P[3], *find_if(all(P), [&](auto p) { return sgn(volume(P
      [0], P[1], P[2], p)) != 0; }));
  vector<vector<int>> flag(n, vector<int>(n));
  res.emplace_back(0, 1, 2); res.emplace_back(2, 1, 0);
  for (int i = 3; i < n; ++i) {
    vector<Face> next;
    for (auto f : res) {
      int d = sgn(volume(P[f.a], P[f.b], P[f.c], P[i]));
      if (d <= 0) next.pb(f);
      int ff = (d > 0) - (d < 0);
      flag[f.a][f.b] = flag[f.b][f.c] = flag[f.c][f.a] = ff;
    }
    for (auto f : res) {
      auto F = [&](int x, int y) {
        if (flag[x][y] > 0 && flag[y][x] <= 0)
          next.emplace_back(x, y, i);
      };
      F(f.a, f.b); F(f.b, f.c); F(f.c, f.a);
    }
    res = next;
  }
}
bool same(Face s, Face t) {
  if (sgn(volume(P[s.a], P[s.b], P[s.c], P[t.a])) != 0) return
      0;
  if (sgn(volume(P[s.a], P[s.b], P[s.c], P[t.b])) != 0) return
      0;
  if (sgn(volume(P[s.a], P[s.b], P[s.c], P[t.c])) != 0) return
      0;
  return 1;
}
int polygon_face_num() {
  int ans = 0;
  for (int i = 0; i < sz(res); ++i)
    ans += none_of(res.begin(), res.begin() + i, [&](Face g) {
        return same(res[i], g); });
  return ans;
}
double get_volume() {
  double ans = 0;
  for (auto f : res)
    ans += volume(Point(0, 0, 0), P[f.a], P[f.b], P[f.c]);
  return fabs(ans / 6);
}
double get_dis(Point p, Face f) {
  Point p1 = P[f.a], p2 = P[f.b], p3 = P[f.c];
  double a = (p2.y - p1.y) * (p3.z - p1.z) - (p2.z - p1.z) * (
      p3.y - p1.y);
  double b = (p2.z - p1.z) * (p3.x - p1.x) - (p2.x - p1.x) * (
      p3.z - p1.z);
  double c = (p2.x - p1.x) * (p3.y - p1.y) - (p2.y - p1.y) * (
      p3.x - p1.x);
  double d = 0 - (a * p1.x + b * p1.y + c * p1.z);
  return fabs(a * p.x + b * p.y + c * p.z + d) / sqrt(a * a + b
      * b + c * c);
}
};
// n^2 delaunay: facets with negative z normal of
// convexhull of (x, y, x^2 + y^2), use a pseudo-point
// (0, 0, inf) to avoid degenerate case
```

DelaunayTriangulation.cpp

Description: Constructs an undirected Delaunay triangulation of 2D integer points by divide and conquer. No point may lie strictly inside a resulting circumcircle.

Time: $\mathcal{O}\left(N^2\right)$

1131c7, 60 lines

```
/* Delaunay Triangulation:
   Given a sets of points on 2D plane, find a
   triangulation such that no points will strictly
   inside circumcircle of any triangle. */
struct Edge {
  int id; // oidx[id]
  list<Edge>::iterator twin;
  Edge(int _id = 0):id(_id) {}
};
struct Delaunay { // 0-base
  int n;
  vector<int> oidx;
  vector<list<Edge>> head; // result udir. graph
  vector<pll> p;
  Delaunay(int _n, vector<pll> _p): n(_n), oidx(n), head(n), p(
      n) {
    iota(all(oidx), 0);
    for (int i = 0; i < n; ++i) head[i].clear();
    sort(all(oidx), [&](int a, int b)
        { return _p[a] < _p[b]; });
    for (int i = 0; i < n; ++i) p[i] = _p[oidx[i]];
    divide(0, n - 1);
  }
  void addEdge(int u, int v) {
    head[u].push_front(Edge(v));
    head[v].push_front(Edge(u));
    head[u].begin()->twin = head[v].begin();
    head[v].begin()->twin = head[u].begin();
  }
  void divide(int l, int r) {
    if (l == r) return;
    if (l + 1 == r) return addEdge(l, l + 1);
    int mid = (l + r) >> 1, nw[2] = {l, r};
    divide(l, mid), divide(mid + 1, r);
    auto gao = [&](int t) {
      pll pt[2] = {p[nw[0]], p[nw[1]]};
      for (auto it : head[nw[t]]) {
        int v = ori(pt[1], pt[0], p[it.id]);
        if (v > 0 || (v == 0 && abs2(pt[t ^ 1] - p[it.id]) <
            abs2(pt[1] - pt[0])))
          return nw[t] = it.id, true;
      }
      return false;
    };
    while (gao(0) || gao(1));
    addEdge(nw[0], nw[1]); // add tangent
    while (true) {
      pll pt[2] = {p[nw[0]], p[nw[1]]};
      int ch = -1, sd = 0;
      for (int t = 0; t < 2; ++t)
        for (auto it : head[nw[t]])
          if (ori(pt[0], pt[1], p[it.id]) > 0 && (ch == -1 ||
              in_cc({pt[0], pt[1], p[ch]}, p[it.id])))
            ch = it.id, sd = t;
      if (ch == -1) break; // upper common tangent
      for (auto it = head[nw[sd]].begin(); it != head[nw[sd]].
          end(); )
        if (seg_strict_intersect(pt[sd], p[it->id], pt[sd ^ 1],
            p[ch]))
          head[it->id].erase(it->twin), head[nw[sd]].erase(it
              ++);
        else ++it;
      nw[sd] = ch, addEdge(nw[0], nw[1]);
    }
  }
};
```

VoronoiDiagram.cpp

Description: Builds the perpendicular-bisector half-planes of every Voronoi cell from a Delaunay triangulation. Run half-plane intersection to obtain the actual cells.

Time: $\mathcal{O}\left(N^2\right)$ including Delaunay construction

e4f408, 12 lines

```
// all coord. is even, you may want to call halfPlaneInter
    after then
vector<vector<Line>> vec;
void build_voronoi_line(int n, vector<pll> &pts) {
  Delaunay tool(n, pts); // Delaunay
  vec.clear(), vec.resize(n);
  for (int i = 0; i < n; ++i)
    for (auto e : tool.head[i]) {
      int u = tool.oidx[i], v = tool.oidx[e.id];
      pll m = (pts[v] + pts[u]) / 2LL, d = perp(pts[v] - pts[u
          ]);
      vec[u].pb(Line(m, m + d));
    }
}
```

PolyUnion.cpp

Description: Computes the area of the union of counter-clockwise integer polygons by splitting each edge into covered and uncovered parameter intervals.

Time: $\mathcal{O}\left(E^2 \log E\right)$, where E is the total number of polygon edges

a9e419, 36 lines

```
ld rat(pll a, pll b) {
  return sgn(b.X) ? (ld)a.X / b.X : (ld)a.Y / b.Y;
} // all poly. should be ccw
ld polyUnion(vector<vector<pll>> &poly) {
  ld res = 0;
  for (auto &p : poly)
    for (int a = 0; a < sz(p); ++a) {
      pll A = p[a], B = p[(a + 1) % sz(p)];
      vector<pair<ld, int>> segs = {{0, 0}, {1, 0}};
      for (auto &q : poly) {
        if (&p == &q) continue;
        for (int b = 0; b < sz(q); ++b) {
          pll C = q[b], D = q[(b + 1) % sz(q)];
          int sc = ori(A, B, C), sd = ori(A, B, D);
          if (sc != sd && min(sc, sd) < 0) {
            ld sa = cross(D - C, A - C), sb = cross(D - C, B -
                C);
            segs.pb({sa / (sa - sb), sgn(sc - sd)});
          }
          if (!sc && !sd && &q < &p && sgn(dot(B - A, D - C)) >
              0) {
            segs.pb({rat(C - A, B - A), 1});
            segs.pb({rat(D - A, B - A), -1});
          }
        }
      }
      sort(all(segs));
      for (auto &s : segs) s.X = clamp(s.X, 0.0, 1.0);
      ld sum = 0;
      int cnt = segs[0].second;
      for (int j = 1; j < sz(segs); ++j) {
        if (!cnt) sum += segs[j].X - segs[j - 1].X;
        cnt += segs[j].Y;
      }
      res += cross(A, B) * sum;
    }
  return res / 2;
}
```

TangentPointToHull.cpp

Description: Finds one contact vertex for each tangent from a point strictly outside a convex polygon. The polygon must be stored in cyclic order.
Time: $\mathcal{O}(\log N)$

8a5cef, 9 lines

```
/* The point should be strictly out of hull
   return arbitrary point on the tangent line */
pii get_tangent(vector<pll> &C, pll p) {
  auto gao = [&](int s) {
    return cyc_tsearch(sz(C), [&](int x, int y)
    { return ori(p, C[x], C[y]) == s; });
  };
  return pii(gao(1), gao(-1));
} // return (a, b), ori(p, C[a], C[b]) >= 0
```

Heart.cpp

Description: Computes the circumcenter, incenter, centroid, and orthocenter of a non-degenerate triangle.
Time: $\mathcal{O}(1)$

df5857, 18 lines

```
pdd circenter(pdd p0, pdd p1, pdd p2) { // SCOPE HASH
  p1 = p1 - p0, p2 = p2 - p0; // radius = abs(center)
  double x1 = p1.X, y1 = p1.Y, x2 = p2.X, y2 = p2.Y;
  double m = 2. * (x1 * y2 - y1 * x2);
  pdd center;
  center.X = (x1 * x1 * y2 - x2 * x2 * y1 + y1 * y2 * (y1 - y2)
      ) / m;
  center.Y = (x1 * x2 * (x2 - x1) - y1 * y1 * x2 + x1 * y2 * y2
      ) / m;
  return center + p0;
}
pdd incenter(pdd p1, pdd p2, pdd p3) { // radius = area / s * 2
  double a = abs(p2 - p3), b = abs(p1 - p3), c = abs(p1 - p2);
  double s = a + b + c;
  return (a * p1 + b * p2 + c * p3) / s;
}
pdd masscenter(pdd p1, pdd p2, pdd p3)
{ return (p1 + p2 + p3) / 3; }
pdd orthcenter(pdd p1, pdd p2, pdd p3)
{ return masscenter(p1, p2, p3) * 3.0 - circenter(p1, p2, p3) *
     2.0; }
```

RotatingSweepLine.cpp

Description: Rotates a sweep direction through all pairwise point directions while maintaining the order of points and grouping simultaneous collinear events.
Time: $\mathcal{O}\left(N^2 \log N\right)$

eacb36, 31 lines

```
struct Event {
  pll d; int u, v;
  bool operator<(const Event &b) const {
    int ret = cmp(d, b.d, false);
    return ret == -1 ? false : ret; } // no tie-break
};
void rotatingSweepLine(const vector<pll> &p) {
  const int n = sz(p);
  vector<Event> e; e.reserve(n * (n - 1));
  for (int i = 0; i < n; i++)
    for (int j = 0; j < n; j++) // pos[i] < pos[j] when the
        event occurs
      if (i != j) e.pb({p[j] - p[i], i, j});
  sort(all(e));
  vector<int> ord(n), pos(n);
  iota(all(ord), 0);
  sort(all(ord), [&](int i, int j) { // initial order
      return p[i].Y != p[j].Y ? p[i].Y < p[j].Y : p[i].X < p[j
          ].X; });
  for (int i = 0; i < n; i++) pos[ord[i]] = i;
  // initialize
  for (int i = 0, j = 0; i < sz(e); i = j) {
    // do something
    vector<pii> tmp;
    for (; j < sz(e) && !(e[i] < e[j]); j++)
      tmp.pb(pii(e[j].u, e[j].v));
    sort(all(tmp), [&](pii x, pii y){
        return pii(pos[x.F], pos[x.S]) < pii(pos[y.F], pos[y.S
            ]); });
    for (auto [x, y] : tmp) // pos[x] + 1 == pos[y]
      tie(ord[pos[x]], ord[pos[y]], pos[x], pos[y]) =
        make_tuple(ord[pos[y]], ord[pos[x]], pos[y], pos[x]);
  }
}
```

VectorInPoly.cpp

Description: Tests whether a directed vector from a polygon vertex lies inside the local angle formed by the previous and next vertices, with optional strict boundaries.
Time: $\mathcal{O}(1)$

c6d0fa, 10 lines

```
// ori(a, b, c) >= 0, valid: "strict" angle from a-b to a-c
bool btwangle(pll a, pll b, pll c, pll p, int strict) {
  return ori(a, b, p) >= strict && ori(a, p, c) >= strict;
}
// whether vector{cur, p} in counter-clockwise order prv, cur,
    nxt
bool inside(pll prv, pll cur, pll nxt, pll p, int strict) {
  if (ori(cur, nxt, prv) >= 0)
    return btwangle(cur, nxt, prv, p, strict);
  return !btwangle(cur, prv, nxt, p, !strict);
}
```

ConvexHullDP.cpp

Description: Orders directed point pairs by angle and provides an $O(N^3)$ transition skeleton for dynamic programming over convex polygons.
Time: $\mathcal{O}\left(N^3\right)$

ec5414, 33 lines

```
sort(all(pts), [&](pll x, pll y) {
  return x.Y != y.Y ? x.Y < y.Y : x.X < y.X;
});
auto getvec = [&](pii x) { return pts[x.S] - pts[x.F]; };
vector<pii> trans;
for (int j = 0; j < n; j++)
  for (int k = 0; k < n; k++)
    if (j != k) trans.pb(pii(j, k));
sort(all(trans), [&](pii x, pii y) -> bool{
  int tmp = cmp(getvec(x), getvec(y), false);
  if (tmp != -1) return tmp;
  pll v = getvec(x);
  return dot(v, pts[x.F]) > dot(v, pts[y.F]);
});
// DP for convex hull vertices (no points on edges)
auto solve = [&](int bottom) { // O(n^3)
  // vector<ll> dp(n);
  for (int j = bottom + 1; j < n; j++) {
    // check whether bottom -> j is legal
    // init trans -> j
  }
  for (auto [i, j] : trans) {
    if (i <= bottom || j <= bottom ||
        ori(pts[bottom], pts[i], pts[j]) <= 0) continue;
    // check whether i -> j is legal
    // normal trans i -> j
  }
  for (int j = bottom + 1; j < n; j++) {
    // check whether j -> bottom is legal
    // end trans j ->
  }
};
for(int i = 0; i < n; i++) solve(i);
```

CountPointsInTriangle.cpp

Description: Number of points of B strictly inside triangle A[a], A[b], A[c] (points on the border or equal to a vertex do not count). A = B is fine.
Usage: `TriCount tc(A, B); tc.query(a, b, c);`
Time: $\mathcal{O}\left(N^2 M\right)$ preprocessing, $\mathcal{O}(1)$ per query

93b23c, 25 lines

```
struct TriCount {
  vector<pll> A, B; vector<vector<int32_t>> lo, on; vi eq;
  TriCount(const vector<pll>& a, vector<pll> b) : A(a), B(b),
      lo(sz(a), vector<int32_t>(sz(a))), on(lo), eq(sz(a)) {
    sort(all(B));
    rep(i, 0, sz(A)){
      int s = upper_bound(all(B), A[i]) - B.begin();
      eq[i] = s - (lower_bound(all(B), A[i]) - B.begin());
      rep(j, 0, sz(A)) if(A[i] < A[j])
        for(int k = s; k < sz(B) && B[k] < A[j]; k++){
          int o = ori(A[i], A[j], B[k]); // lo: strictly below,
              on: on the open segment
          lo[i][j] += o < 0, on[i][j] += o == 0;
        }
    }
  }
  int query(int a, int b, int c){
    if(A[b] < A[a]) swap(a, b);
    if(A[c] < A[b]) swap(b, c);
    if(A[b] < A[a]) swap(a, b);
    int o = ori(A[a], A[c], A[b]);
    if(o > 0) return lo[a][b] + lo[b][c] - lo[a][c] - on[a][c];
    if(o < 0) return lo[a][c] - lo[a][b] - on[a][b] - lo[b][c]
        - on[b][c] - eq[b];
    return 0;
  }
};
```

ClosestPair.cpp

Description: Indices of a closest pair of points ($n \geq 2$). Needs Geometry-Template.
Time: $\mathcal{O}(n \log n)$

90f3ea, 15 lines

```
pii closestPair(const vector<pll>& v){
  int n = sz(v), j = 0; vi id(n); iota(all(id), 0);
  sort(all(id), [&](int a, int b){ return v[a].Y < v[b].Y; });
  set<pair<pll, int>> S; pair<ll, pii> ret{LLONG_MAX, {0, 1}};
  for(int i : id){
    pll p = v[i]; ll d = 1 + (ll)sqrtl(ret.F);
    while(v[id[j]].Y <= p.Y - d) S.erase({v[id[j]], id[j]}), j
        ++;
    auto lo = S.lower_bound({{p.X - d, LLONG_MIN}, -1});
    auto hi = S.upper_bound({{p.X + d, LLONG_MAX}, n});
    for(; lo != hi; ++lo) ret = min(ret, {abs2(lo->F - p), {lo
        ->S, i}});
    if(!ret.F) break; // else O(n^2) on equal points
    S.insert({p, i});
  }
  return ret.S;
}
```

Strings (9)

Hashing.h

Description: Self-explanatory methods for string hashing.

3fdef6, 44 lines

```
// Arithmetic mod 2^64-1. 2x slower than mod 2^64 and more
// code, but works on evil test data (e.g. Thue-Morse, where
// ABBA... and BAAB... of length 2^10 hash the same mod 2^64).
// "typedef ull H;" instead if you think test data is random,
// or work mod 10^9+7 if the Birthday paradox is not a problem.
typedef uint64_t ull;
```

```
struct H {
  ull x; H(ull x=0) : x(x){}
  H operator+(H o){ return x + o.x + (x + o.x < x); }
  H operator-(H o){ return *this + ~o.x; }
  H operator*(H o){ auto m = (__uint128_t)x * o.x;
    return H((ull)m) + (ull)(m >> 64); }
  ull get() const { return x + !~x; }
  bool operator==(H o) const { return get() == o.get(); }
  bool operator<(H o) const { return get() < o.get(); }
};
static const H C = (ll)1e11+3; // (order ~ 3e9; random also ok)

struct HashInterval {
  vector<H> ha, pw;
  HashInterval(string& str) : ha(sz(str)+1), pw(ha){
    pw[0] = 1;
    rep(i, 0, sz(str))
      ha[i+1] = ha[i] * C + str[i],
      pw[i+1] = pw[i] * C;
  }
  H hashInterval(int a, int b){ // hash [a, b)
    return ha[b] - ha[a] * pw[b - a];
  }
};

H hashString(string& s){H h{}; for(char c:s) h=h*C+c;return h;}
      // PART HASH = d099fb

vector<H> getHashes(string& str, int length){
  if(sz(str) < length) return {};
  H h = 0, pw = 1;
  rep(i, 0, length)
    h = h * C + str[i], pw = pw * C;
  vector<H> ret = {h};
  rep(i, length, sz(str)){
    ret.push_back(h = h * C + str[i] - pw * str[i-length]);
  }
  return ret;
} // FULL HASH = 3fdef6
```

KMP.h

Description: pi[x] computes the length of the longest prefix of s that ends at x, other than s[0...x] itself (abacaba -> 0010123). Can be used to find all occurrences of a string.
Time: $\mathcal{O}(n)$ d4375c, 16 lines

```
vi pi(const string& s){
  vi p(sz(s));
  rep(i, 1, sz(s)){
    int g = p[i-1];
    while(g && s[i] != s[g]) g = p[g-1];
    p[i] = g + (s[i] == s[g]);
  }
  return p;
}

vi match(const string& s, const string& pat){
  vi p = pi(pat + '\0' + s), res;
  rep(i, sz(p)-sz(s), sz(p))
    if(p[i] == sz(pat)) res.push_back(i - 2 * sz(pat));
  return res;
}
```

Zfunc.h

Description: z[i] computes the length of the longest common prefix of s[i:] and s, except z[0] = 0. (abacaba -> 0010301)
Time: $\mathcal{O}(n)$ ee09e2, 12 lines

```
vi Z(const string& S){
  vi z(sz(S));
  int l = -1, r = -1;
  rep(i, 1, sz(S)){
    z[i] = i >= r ? 0 : min(r - i, z[i - l]);
    while(i + z[i] < sz(S) && S[i + z[i]] == S[z[i]])
      z[i]++;
    if(i + z[i] > r)
      l = i, r = i + z[i];
  }
  return z;
}
```

Manacher.h

Description: For each position in a string, computes p[0][i] = half length of longest even palindrome around pos i, p[1][i] = longest odd (half rounded down).
Time: $\mathcal{O}(N)$ e7ad79, 13 lines

```
array<vi, 2> manacher(const string& s){
  int n = sz(s);
  array<vi, 2> p = {vi(n+1), vi(n)};
  rep(z, 0, 2) for(int i=0, l=0, r=0; i < n; i++){
    int t = r-i+!z;
    if(i<r) p[z][i] = min(t, p[z][l+t]);
    int L = i-p[z][i], R = i+p[z][i]-!z;
    while(L>=1 && R+1<n && s[L-1] == s[R+1])
      p[z][i]++, L--, R++;
    if(R>r) l=L, r=R;
  }
  return p;
}
```

Eertree.h

Description: Palindromic tree. Node 0 is the odd root (len -1), node 1 the even root (len 0); other nodes are the distinct palindromes in order of first occurrence. After add(c), last is the longest palindromic suffix. count() makes cnt[v] = number of occurrences. For $n = 10^6$ call reserve(n + 2) on the vectors first.
Time: $\mathcal{O}(n)$ amortized, $\mathcal{O}(n\sigma)$ memory a0c393, 22 lines

```
struct Eertree {
  vector<array<int32_t, 26>> nx; vector<int32_t> len, link, par
      , cnt;
  string s; int last = 1;
  Eertree(){ node(-1, 0, -1), node(0, 0, -1); }
  int node(int l, int k, int p){
    nx.push_back({}), len.push_back(l), link.push_back(k);
    par.push_back(p), cnt.push_back(0); return sz(len) - 1;
  }
  int get(int v, int i){
    while(i - len[v] < 1 || s[i - len[v] - 1] != s[i]) v = link
        [v];
    return v;
  }
  void add(char ch){
    s += ch; int i = sz(s) - 1, c = ch - 'a', v = get(last, i);
    if(!nx[v][c]){
      int k = v ? nx[get(link[v], i)][c] : 1;
      int u = node(len[v] + 2, k, v); nx[v][c] = u;
    }
    cnt[last = nx[v][c]]++;
  }
  void count(){ for(int v = sz(len) - 1; v > 1; v--) cnt[link[v
      ]] += cnt[v]; }
};
```

MinRotation.h

Description: Finds the lexicographically smallest rotation of a string.
Usage: `rotate(v.begin(), v.begin()+minRotation(v), v.end());`
Time: $\mathcal{O}(N)$ 67652e, 8 lines

```
int minRotation(string s){
  int a=0, N=sz(s); s += s;
  rep(b, 0, N) rep(k, 0, N){
    if(a+k == b || s[a+k] < s[b+k]){b += max<int>(0, k-1);
        break;}
    if(s[a+k] > s[b+k]){ a = b; break; }
  }
  return a;
}
```

BitsetLCS.h

Description: Length of the longest common subsequence, 64 columns per word operation.
Time: $\mathcal{O}(|a||b|/64)$, $\mathcal{O}(256|b|/64)$ memory ceb242, 17 lines

```
typedef unsigned long long ull;
int lcs(const string& a, const string& b){
  int W = sz(b) / 64 + 1, res = sz(b);
  vector<vector<ull>> M(256, vector<ull>(W));
  rep(j, 0, sz(b)) M[(unsigned char)b[j]][j >> 6] |= 1ULL << (j
      & 63);
  vector<ull> V(W, ~0ULL); // V = (V + (V & M[c])) | (V & ~M[c
      ])
  for(unsigned char c : a){
    unsigned __int128 carry = 0;
    rep(w, 0, W){
      ull u = V[w] & M[c][w];
      carry += (unsigned __int128)V[w] + u;
      V[w] = (ull)carry | (V[w] - u), carry >>= 64;
    }
  }
  rep(j, 0, sz(b)) res -= V[j >> 6] >> (j & 63) & 1;
  return res;
}
```

SuffixArray.h

Description: Builds suffix array for a string. `sa[i]` is the starting index of the suffix which is i'th in the sorted suffix array. The returned vector is of size $n + 1$, and `sa[0] = n`. The `lcp` array contains longest common prefixes for neighbouring strings in the suffix array: `lcp[i] = lcp(sa[i], sa[i-1])`, `lcp[0] = 0`. The input string must not contain any nul chars.
Time: $\mathcal{O}(n \log n)$ ed2029, 26 lines

```
struct SuffixArray {
  vi sa, lcp;
  SuffixArray(string s, int lim=256){ // or vector<int>
    s.push_back(0); int n = sz(s), k = 0, a, b;
    vi x(n), y(n), ws(max(n, lim));
    rep(i, 0, n) x[i] = (unsigned char)s[i];
    sa = lcp = y, iota(all(sa), 0);
    for(int j = 0, p = 0; p < n; j = max<int>(1, j * 2), lim =
        p){
      p = j, iota(all(y), n - j);
      rep(i, 0, n) if(sa[i] >= j) y[p++] = sa[i] - j;
      fill(all(ws), 0);
      rep(i, 0, n) ws[x[i]]++;
      rep(i, 1, lim) ws[i] += ws[i - 1];
      for(int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
      swap(x, y), p = 1, x[sa[0]] = 0;
      rep(i, 1, n) a = sa[i - 1], b = sa[i], x[b] =
        (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++;
    }
    for(int i = 0, j; i < n - 1; i++){
      if(k) k--;
      j = sa[x[i] - 1];
      while(s[i + k] == s[j + k]) k++;
      lcp[x[i]] = k;
    }
```

```
  }
};
```

SuffixArraySAIS.h

Description: Linear suffix array by SA-IS. Includes the empty suffix.
Time: $\mathcal{O}(n)$

e66dbd, 66 lines

```
struct SAIS {
  vi sa, lcp;
  vi build(const vi &s, int lim){
    int n = sz(s);
    if(n == 1) return {0};
    vi ret(n, -1), ls(n), sumL(lim + 1), sumS(lim + 1);
    for(int i = n - 2; i >= 0; i--){
      ls[i] = s[i] == s[i + 1] ? ls[i + 1] : s[i] < s[i + 1];
    }
    rep(i, 0, n) (ls[i] ? sumL[s[i] + 1] : sumS[s[i]])++;
    rep(i, 0, lim + 1){
      sumS[i] += sumL[i];
      if(i < lim) sumL[i + 1] += sumS[i];
    }
    auto induce = [&](const vi &lms){
      fill(all(ret), -1);
      vi buf = sumS;
      for(int x : lms) ret[buf[s[x]]++] = x;
      buf = sumL, ret[buf[s[n - 1]]++] = n - 1;
      rep(i, 0, n){
        int x = ret[i] - 1;
        if(x >= 0 && !ls[x]) ret[buf[s[x]]++] = x;
      }
      buf = sumL;
      for(int i = n; i--;){
        int x = ret[i] - 1;
        if(x >= 0 && ls[x]) ret[--buf[s[x] + 1]] = x;
      }
    };
    vi id(n + 1, -1), lms;
    rep(i, 1, n) if(!ls[i - 1] && ls[i]) id[i] = sz(lms), lms.
         pb(i);
    induce(lms);
    if(sz(lms)){
      vi ord;
      for(int x : ret) if(id[x] != -1) ord.pb(x);
      vi rec(sz(lms)); int names = 0;
      rec[id[ord[0]]] = 0;
      rep(i, 1, sz(lms)){
        int l = ord[i - 1], r = ord[i];
        int el = id[l] + 1 < sz(lms) ? lms[id[l] + 1] : n;
        int er = id[r] + 1 < sz(lms) ? lms[id[r] + 1] : n;
        bool same = el - l == er - r;
        while(same && l < el && s[l] == s[r]) l++, r++;
        if(l == n || s[l] != s[r]) same = false;
        rec[id[ord[i]]] = same ? names : ++names;
      }
      vi recSa = build(rec, names);
      rep(i, 0, sz(lms)) ord[i] = lms[recSa[i]];
      induce(ord);
    }
    return ret;
  }
  SAIS(string s){
    int n = sz(s), k = 0;
    vi a(n + 1), rk(n + 1);
    rep(i, 0, n) a[i] = (unsigned char)s[i];
    sa = build(a, 255), lcp.assign(n + 1, 0);
    rep(i, 0, n + 1) rk[sa[i]] = i;
    rep(i, 0, n){
      if(k) k--;
      int j = sa[rk[i] - 1];
      while(a[i + k] == a[j + k]) k++;
      lcp[rk[i]] = k;
    }
  }
};
```

AhoCorasick.h

Description: Aho-Corasick automaton, used for multiple pattern matching. Initialize with AhoCorasick ac(patterns); the automaton start node will be at index 0. find(word) returns for each position the index of the longest word that ends there, or -1 if none. findAll(−, word) finds all words (up to $N\sqrt{N}$ many if no duplicate patterns) that start at each position (shortest first). Duplicate patterns are allowed; empty patterns are not. To find the longest words that start at each position, reverse all input. For large alphabets, split each symbol into chunks, with sentinel bits for symbol boundaries.
Time: construction takes $\mathcal{O}(26N)$, where N = sum of length of patterns. find(x) is $\mathcal{O}(N)$, where N = length of x. findAll is $\mathcal{O}(NM)$.

4697a4, 67 lines

```
struct AhoCorasick {
  enum {alpha = 26, first = 'A'}; // change this!
  struct Node {
    // (nmatches is optional)
    int back, start = -1, end = -1, nmatches = 0;
    signed next[alpha]; // not ll: memory
    Node(int v){ memset(next, v, sizeof(next)); }
  };
  vector<Node> N;
  vi backp;
  void insert(string& s, int j){
    assert(!s.empty());
    int n = 0;
    for(char c : s){
      signed& m = N[n].next[c - first];
      if(m == -1){ n = m = sz(N); N.emplace_back(-1); }
      else n = m;
    }
    if(N[n].end == -1) N[n].start = j;
    backp.push_back(N[n].end);
    N[n].end = j;
    N[n].nmatches++;
  }
  AhoCorasick(vector<string>& pat) : N(1, -1){
    rep(i, 0, sz(pat)) insert(pat[i], i);
    N[0].back = sz(N);
    N.emplace_back(0);

    queue<int> q;
    for(q.push(0); !q.empty(); q.pop()){
      int n = q.front(), prev = N[n].back;
      rep(i, 0, alpha){
        signed &ed = N[n].next[i], y = N[prev].next[i];
        if(ed == -1) ed = y;
        else {
          N[ed].back = y;
          (N[ed].end == -1 ? N[ed].end : backp[N[ed].start])
            = N[y].end;
          N[ed].nmatches += N[y].nmatches;
          q.push(ed);
        }
      }
    }
  }
  vi find(string word){
    int n = 0;
    vi res; // ll count = 0;
    for(char c : word){
      n = N[n].next[c - first];
      res.push_back(N[n].end);
      // count += N[n].nmatches;
    }
    return res;
  }
  vector<vi> findAll(vector<string>& pat, string word){
    vi r = find(word);
    vector<vi> res(sz(word));
    rep(i, 0, sz(word)){
      int ind = r[i];
      while(ind != -1){
        res[i - sz(pat[ind]) + 1].push_back(ind);
        ind = backp[ind];
      }
    }
    return res;
  }
};
```

Sam.h

Description: Generalized suffix automaton for lowercase strings. Construct with total length, insert strings, build, then solve for cnt.
Time: $\mathcal{O}(Nalphabet)$

efe9d5, 38 lines

```
struct exSAM{
  static const int C = 27;
  vector<int>len, link, ord, cnt;
  vector<array<signed, C>>go; // not ll: memory
  int last;
  exSAM(int n){
    len.reserve(2 * n + 2), link.reserve(2 * n + 2), cnt.
        reserve(2 * n + 2), go.reserve(2 * n + 2);
    node(); link[0] = -1, last = 0;
  }
  int node(){
    len.pb(0), link.pb(0), cnt.pb(0), go.pb({}); return sz(len)
        - 1;
  }
  void extend(int c){
    int cur = node(), p = last; len[cur] = len[last] + 1, cnt[
        cur] = 1, last = cur;
    for(; p != -1 and !go[p][c]; p = link[p]) go[p][c] = cur;
    if(p == -1) return link[cur] = 0, void();
    int q = go[p][c];
    if(len[p] + 1 == len[q]) return link[cur] = q, void();
    int cl = node(); go[cl] = go[q], len[cl] = len[p] + 1, link
        [cl] = link[q];
    for(; p != -1 and go[p][c] == q; p = link[p]) go[p][c] = cl
        ;
    link[q] = link[cur] = cl;
  }
  void insert(const string &s){
    for(char c : s) extend(c - 'a');
    extend(26);
  }
  void build(){
    int n = sz(len); ord.resize(n); vector<int>sum(len[last] +
        1);
    rep(i, 0, n) ++sum[len[i]];
    partial_sum(all(sum), sum.begin());
    rep(i, 0, n) ord[--sum[len[i]]] = i;
  }
  void solve(){
    for(int i = sz(ord) - 1; i; --i){
      int u = ord[i]; cnt[link[u]] += cnt[u];
    }
  }
};
```

Various (10)

10.1 Intervals

IntervalContainer.h

Description: Add and remove intervals from a set of disjoint intervals. Will merge the added interval with any overlapping intervals in the set when adding. Intervals are [inclusive, exclusive).
Time: $\mathcal{O}(\log N)$

edce47, 23 lines

```
set<pii>::iterator addInterval(set<pii>& is, int L, int R){
  if(L == R) return is.end();
  auto it = is.lower_bound({L, R}), before = it;
  while(it != is.end() && it->first <= R){
    R = max(R, it->second);
    before = it = is.erase(it);
  }
  if(it != is.begin() && (--it)->second >= L){
    L = min(L, it->first);
    R = max(R, it->second);
    is.erase(it);
  }
  return is.insert(before, {L, R});
}

void removeInterval(set<pii>& is, int L, int R){
  if(L == R) return;
  auto it = addInterval(is, L, R);
  auto r2 = it->second;
  if(it->first == L) is.erase(it);
  else (int&)it->second = L;
  if(R != r2) is.emplace(R, r2);
}
```

IntervalCover.h

Description: Compute indices of smallest set of intervals covering another interval. Intervals should be [inclusive, exclusive). To support [inclusive, inclusive], change (A) to add `|| R.empty()`. Returns empty set on failure (or if G is empty).
Time: $\mathcal{O}(N \log N)$

9e9d8d, 19 lines

```
template<class T>
vi cover(pair<T, T> G, vector<pair<T, T>> I){
  vi S(sz(I)), R;
  iota(all(S), 0);
  sort(all(S), [&](int a, int b){ return I[a] < I[b]; });
  T cur = G.first;
  int at = 0;
  while(cur < G.second){ // (A)
    pair<T, int> mx = make_pair(cur, -1);
    while(at < sz(I) && I[S[at]].first <= cur){
      mx = max(mx, make_pair(I[S[at]].second, S[at]));
      at++;
    }
    if(mx.second == -1) return {};
    cur = mx.first;
    R.push_back(mx.second);
  }
  return R;
}
```

ConstantIntervals.h

Description: Split a monotone function on [from, to) into a minimal set of half-open intervals on which it has the same value. Runs a callback g for each such interval.
Usage: `constantIntervals(0, sz(v), [&](int x){return v[x];}, [&](int lo, int hi, T val){...});`
Time: $\mathcal{O}\left(k \log \frac{n}{k}\right)$

753a4c, 19 lines

```
template<class F, class G, class T>
void rec(int from, int to, F& f, G& g, int& i, T& p, T q){
  if(p == q) return;
  if(from == to){
    g(i, to, p);
    i = to; p = q;
  }else{
    int mid = (from + to) >> 1;
    rec(from, mid, f, g, i, p, f(mid));
    rec(mid+1, to, f, g, i, p, q);
  }
}
template<class F, class G>
void constantIntervals(int from, int to, F f, G g){
  if(to <= from) return;
  int i = from; auto p = f(i), q = f(to-1);
  rec(from, to-1, f, g, i, p, q);
  g(i, to, q);
}
```

10.2 Misc. algorithms

TernarySearch.h

Description: Find the smallest i in $[a, b]$ that maximizes $f(i)$, assuming that $f(a) < \ldots < f(i) \geq \cdots \geq f(b)$. To reverse which of the sides allows non-strict inequalities, change the $<$ marked with (A) to $<=$, and reverse the loop at (B). To minimize f, change it to $>$, also at (B).
Usage: `int ind = ternSearch(0, n-1, [&](int i){return a[i];});`
Time: $\mathcal{O}(\log(b-a))$

9155b4, 11 lines

```
template<class F>
int ternSearch(int a, int b, F f){
  assert(a <= b);
  while(b - a >= 5){
    int mid = (a + b) / 2;
    if(f(mid) < f(mid+1)) a = mid; // (A)
    else b = mid+1;
  }
  rep(i, a+1, b+1) if(f(a) < f(i)) a = i; // (B)
  return a;
}
```

FastKnapsack.h

Description: Given N non-negative integer weights w and a non-negative target t, computes the maximum S <= t such that S is the sum of some subset of the weights.
Time: $\mathcal{O}(N \max(w_i))$

0c7a04, 16 lines

```
int knapsack(vi w, int t){
  int a = 0, b = 0, x;
  while(b < sz(w) && a + w[b] <= t) a += w[b++];
  if(b == sz(w)) return a;
  int m = *max_element(all(w));
  vi u, v(2*m, -1);
  v[a+m-t] = b;
  rep(i, b, sz(w)){
    u = v;
    rep(x, 0, m) v[x+w[i]] = max(v[x+w[i]], u[x]);
    for(x = 2*m; --x > m;) rep(j, max<int>(0, u[x]), v[x])
      v[x-w[j]] = max(v[x-w[j]], j);
  }
  for(a = t; v[a+m-t] < 0; a--) ;
  return a;
}
```

SimulatedAnnealing.h

de78c6, 8 lines

```
double factor = 100000;
const int base = 1e9; // remember to run ~ 10 times
for (int it = 1; it <= 1000000; ++it) {
  // ans: answer, nw: current value, rnd(): mt19937 rnd()
  if (exp(-(nw - ans) / factor) >= (double)(rnd() % base) /
      base)
    ans = nw;
  factor *= 0.99995;
}
```

10.3 Dynamic programming

KnuthDP.h

Description: When doing DP on intervals: $a[i][j] = \min_{i<k<j}(a[i][k] + a[k][j]) + f(i,j)$, where the (minimal) optimal k increases with both i and j, one can solve intervals in increasing order of length, and search $k = p[i][j]$ for $a[i][j]$ only between $p[i][j-1]$ and $p[i+1][j]$. This is known as Knuth DP. Sufficient criteria for this are if $f(b,c) \leq f(a,d)$ and $f(a,c) + f(b,d) \leq f(a,d) + f(b,c)$ for all $a \leq b \leq c \leq d$. Consider also: LineContainer (ch. Data structures), monotone queues, ternary search.
Time: $\mathcal{O}(N^2)$

10.4 Debugging tricks

- `signal(SIGSEGV, [](int) { _Exit(0); });` converts segfaults into Wrong Answers. Similarly one can catch SIGABRT (assertion failures) and SIGFPE (zero divisions). `_GLIBCXX_DEBUG` failures generate SIGABRT (or SIGSEGV on gcc 5.4.0 apparently).
- `feenableexcept(29);` kills the program on NaNs (1), 0-divs (4), infinities (8) and denormals (16).

10.5 Optimization tricks

10.5.1 Bit hacks

- `x & -x` is the least bit in x.
- `for (int x = m; x; ) { --x &= m; ... }` loops over all subset masks of `m` (except `m` itself).
- `c = x&-x, r = x+c; (((r^x) >> 2)/c) | r` is the next number after x with the same number of bits set.
- `rep(b,0,K) rep(i,0,(1 << K)) if (i & 1 << b) D[i] += D[i^(1 << b)];` computes all sums of subsets.

FastMod.h

Description: Compute $a\%b$ about 5 times faster than usual, where b is constant but not known at compile time. Returns a value congruent to a (mod b) in the range $[0, 2b)$.

751a02, 8 lines

```
typedef unsigned long long ull;
struct FastMod {
  ull b, m;
  FastMod(ull b) : b(b), m(-1ULL / b){}
  ull reduce(ull a){ // a % b + (0 or b)
    return a - (ull)((__uint128_t(m) * a) >> 64) * b;
  }
};
```